MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED, LOW-POWER SCHOTTKY TTL, FLIP-FLOPS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, advanced, low-power Schottky TTL, flip-flops, bistable logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual D-type Flip-Flop with Clear and Preset
02	Dual J-$\overline{K}$ Flip-Flop with Clear and Preset
03	Dual J-K Flip-Flop with Clear and Preset
04	Octal D-Type Flip-Flop with 3-State Outputs
05	Octal D-Type Flip-Flop with 3-State Inverted Outputs
06	Dual 4-Bit D-Type Flip-Flop with Clear and 3-State Outputs
07	Dual 4-Bit D-Type Flip-Flop with Preset and 3-State Inverted Outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
K	F-6 (24-lead, 3/8" x 5/8"), flat package
L	D-9 (24-lead, 1/4" x 1-1/4"), dual-in-line package
R	D-8 (20-lead, 1/4" x 1-1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package
3	C-4 (28-terminal, .450" x .450"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.5 V at -18 mA to +7.0 V
Storage temperature	-65°C to +150°C
Maximum power dissipation (P_D) per flip-flop 1/:	
Device types 01 and 02	11 mW
Device type 03	13 mW
Device types 04 and 05	19 mW
Device types 06 and 07	22 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C, D, E, F, K, L, R, and S	(See MIL-M-38510, appendix C)
Cases 2 and 3	80°C/mW 2/
Junction temperature (T_J) 3/	175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.8 V
Case operating temperature range (T_C)	-55°C to +125°C
Width of CLOCK pulse (tp CLOCK):	
Device types 01, 02, 04, 05, 06, and 07	16.5 ns
Device type 03	20 ns
Width of CLEAR pulse (tp $\overline{\text{CLEAR}}$):	
Device types 01, 02, and 03	15 ns
Device type 06	10 ns
Width of PRESET pulse (tp $\overline{\text{PRESET}}$):	
Device types 01, 02, and 03	15 ns
Device type 07	10 ns
Data setup time:	
Device types 01, 02, 04, and 05	15 ns
Device type 03	25 ns
Device types 06 and 07	15 ns
$\overline{\text{CLR}}$ or $\overline{\text{PR}}$ inactive state setup time:	
Device types 01, 02, 05, and 07	10 ns
Device type 03	20 ns
Data hold time (tp HOLD):	
Device type 01	2 ns
Device types 02 and 03	0 ns
Device types 04, 05, 06, and 07	4 ns

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

1/ Must withstand added P_D due to short circuit test (e.g. I_O).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

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STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figures 1 and 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite of qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, except as modified herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V V_{IL} = 0.8 V V_{IH} = 2.0 V	I_{OH} = -400 µA	01,02,03	2.5		V
			I_{OH} = -1.0 mA	04,05, 06,07	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V V_{IL} = 0.8 V V_{IN} = 2.0 V	I_{OL} = 4 mA	01,02,03		0.4	V
			I_{OL} = 12 mA	04,05, 06,07		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, T_C = +25°C	I_{IN} = -18 mA	All		-1.5	V
Low-level input current at CLK or D	I_{IL1}	V_{CC} = 5.5 V V_{IN} = 0.4 V		01	0	-200	µA
Low-level input current at J, K, or $\overline{K}$	I_{IL2}			02,03			
Low-level input current at any input	I_{IL3}			04,05 06,07			
Low-level input current at $\overline{PR}$, $\overline{CLR}$	I_{IL4}	V_{CC} = 5.5 V V_{IL} = 0.4 V		01,02,03	0	-400	µA
High-level input current at CLK or D	I_{IH1}	V_{CC} = 5.5 V V_{IH} = 2.7 V		01		20	µA
High-level input current at J, K, or $\overline{K}$	I_{IH2}			02, 03			
High-level input current at any input	I_{IH3}			04,05, 06,07			
High-level input current at $\overline{PR}$ or $\overline{CLR}$	I_{IH4}			01,02,03		40	µA
High-level input current at CLK	I_{IH5}			02,03		20	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
High-level input current at CLK or D	I_{IH6}	V_{CC} = 5.5 V V_{IN} = 7.0 V		01		100	µA
High-level input current at J, K, or $\overline{K}$	I_{IH7}			02,03		100	µA
High-level input current at any input	I_{IH8}			04,05, 06,07			
High-level input current at $\overline{PR}$ or $\overline{CLR}$	I_{IH9}			01,02,03		200	µA
High-level input current at CLK	I_{IH10}			02		400	µA
				03		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V V_O = 2.25 V		01,02,03	-30	-112	mA
				04,05, 06,07	-15	-110	mA
Output current, high level, outputs OFF	I_{OZH}	V_{CC} = 5.5 V V_O = 2.7 V		04,05, 06,07		20	µA
Output current, low level, outputs OFF	I_{OZL}	V_{CC} = 5.5 V V_O = 0.4 V		04,05, 06,07		-20	µA
Supply current 2/	I_{CC}	V_{CC} = 5.5 V V_{IN} = 0 V		01,02		4.0	mA
				03		4.5	mA
Supply current, outputs high	I_{CCH}	V_{CC} = 5.5 V	V_{IN} = 5.0 V	04,05		17	mA
			V_{IN} = 0 V	06		15	
				07		21	
Supply current, outputs low	I_{CCL}	V_{CC} = 5.5 V	V_{IN} = 0 V	04		23	mA
				06		29	
			V_{IN} = 5.0 V	05		23	
				07		29	
Supply current, outputs disabled	I_{CCZ}	V_{CC} = 5.5 V V_{OC} = 5.0 V		04,05		27	mA
				06,07		31	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Maximum clock frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 500Ω	01,02,04 05,06,07	30		MHz
			03	25		
Propagation delay time low to high level, $\overline{CLR}$ or $\overline{PR}$ to output	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pf ±10% R_L = 500Ω	01,02	3	15	ns
			03	3	20	
Propagation delay time high to low, $\overline{CLR}$ or $\overline{PR}$ to output	t_{PHL1}		01,02	5	17	ns
			03	4	22	
			05,07	6	22	
Propagation delay time low to high level, CLK to output	t_{PLH2}		01,02	5	18	ns
			03	3	18	
			04,05 06,07	4	15	
Propagation delay time high to low level, CLK to output	t_{PHL2}		01,02	5	20	ns
			03	5	23	
			04,05 06,07	4	15	
Propagation delay time output control to low level output	t_{PZL}		04,05 06,07	4	21	ns
Propagation delay time output control to high level output	t_{PZH}		04,05 06,07	4	21	ns
Propagation delay time low level to output control	t_{PLZ}		04,05 06,07	3	15	ns
Propagation delay time high level to output control	t_{PHZ}		04,05 06,07	2	10	ns

1/ The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current, I_{OS}.

2/ I_{CC} is measured with outputs open with J, K, $\overline{K}$, or D, CLK and $\overline{PR}$ grounded; then with J, K, $\overline{K}$, or D, CLK and $\overline{CLR}$ grounded.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for for group C inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

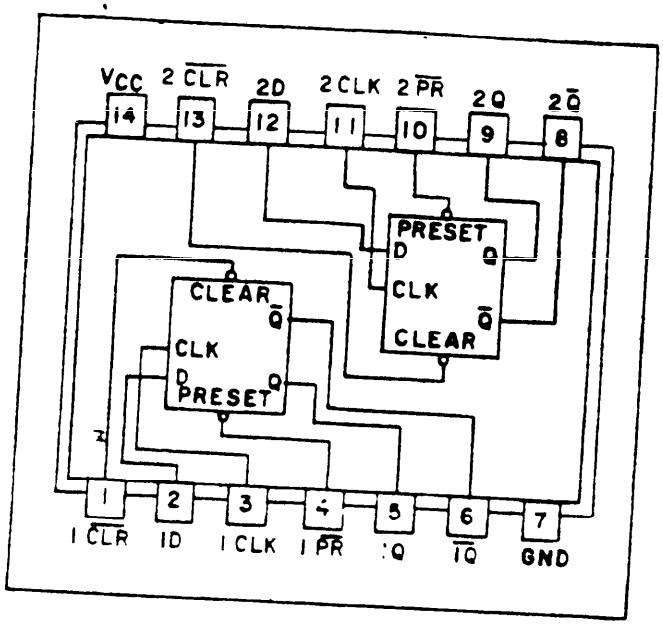
4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

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Device type 01
Cases A, B, C, and D



Device type 01
Case 2

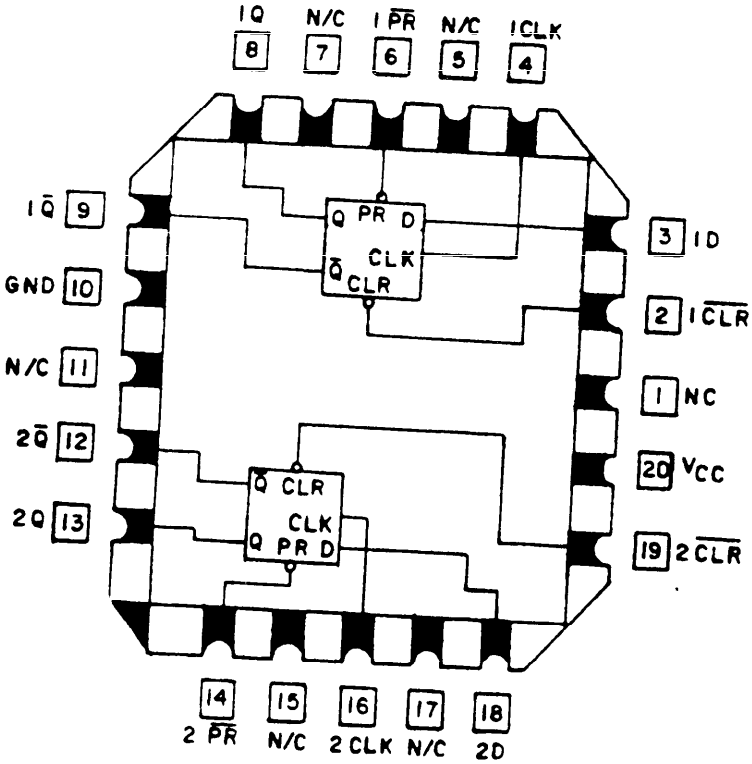
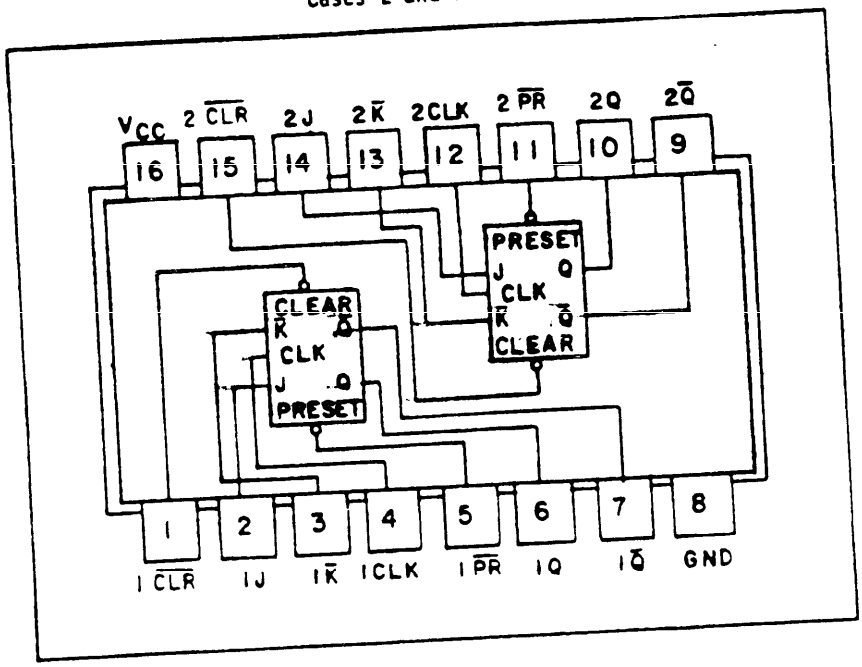


FIGURE 1. Terminal connections (top view).

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Device type 02

Cases E and F



Device type 02

Case 2

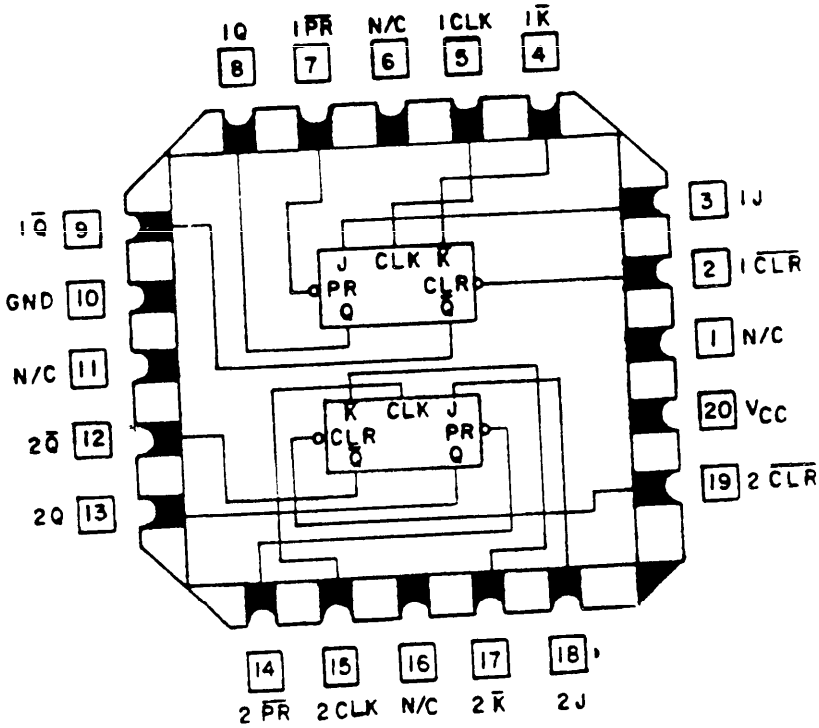
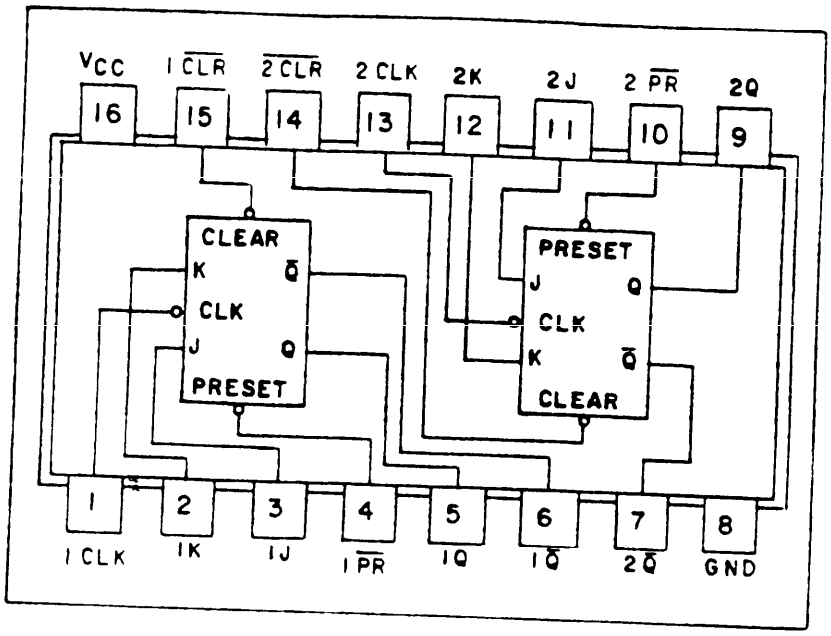


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 03
Cases E and F



Device type 03
Case 2

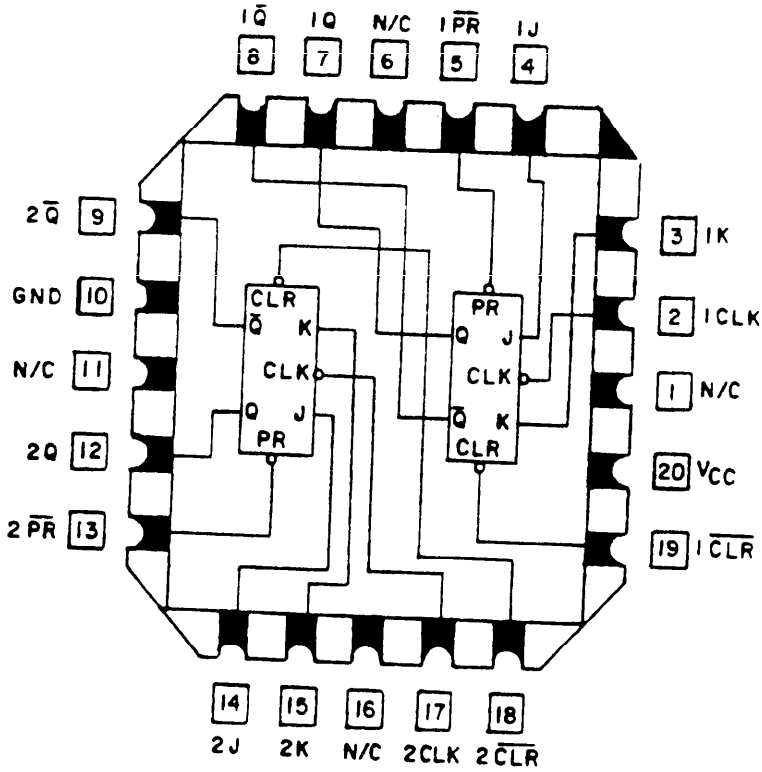
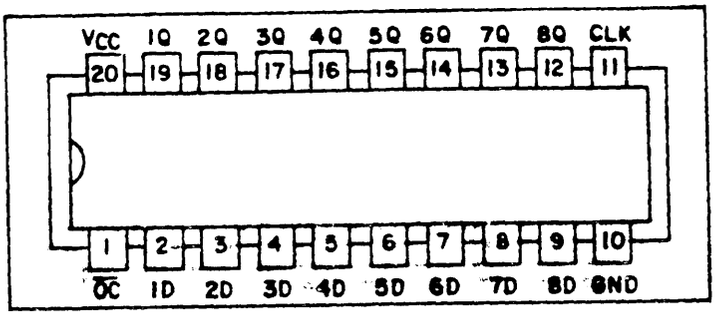


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 04
Cases R and S



Device type 04
Case 2

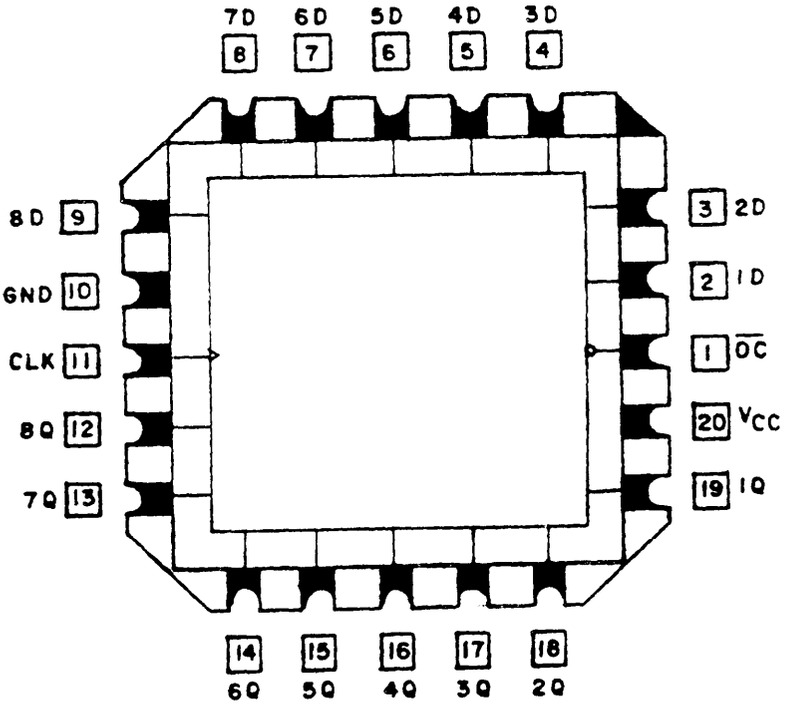
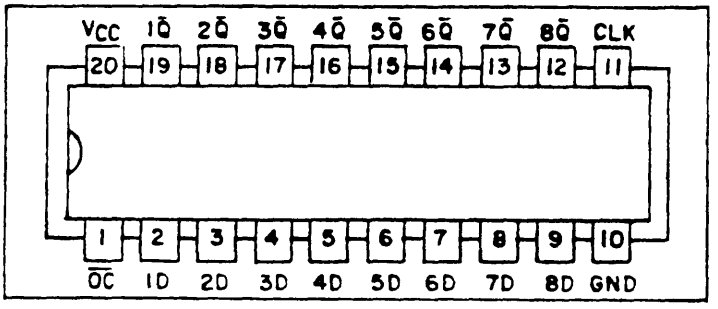


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 05
Cases R and S



Device type 05
Case 2

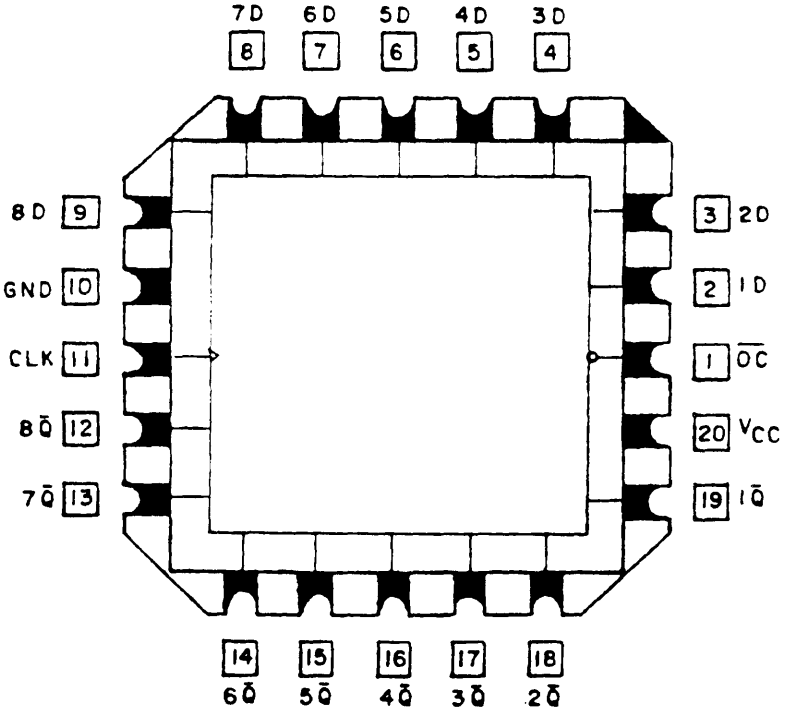
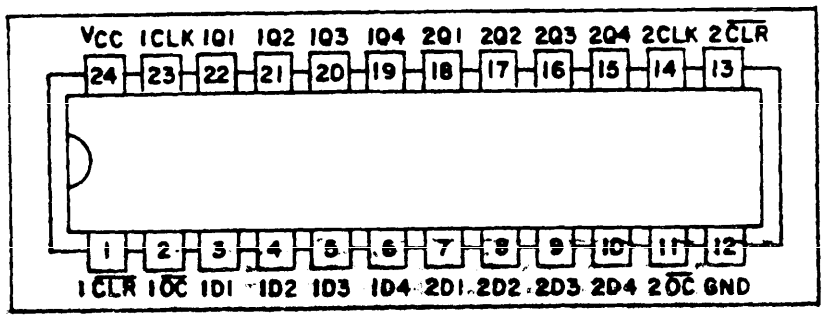


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 06

Cases L and K



Device type 06

Case 3

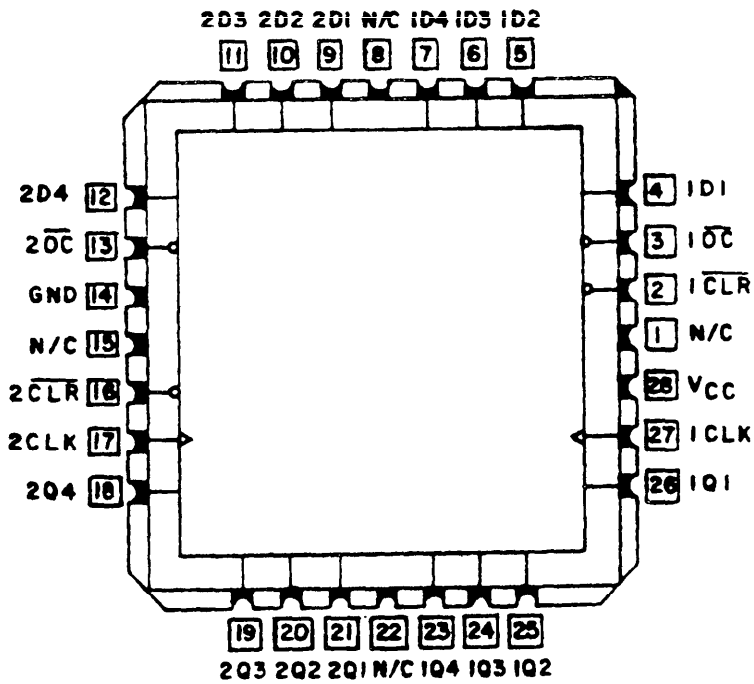
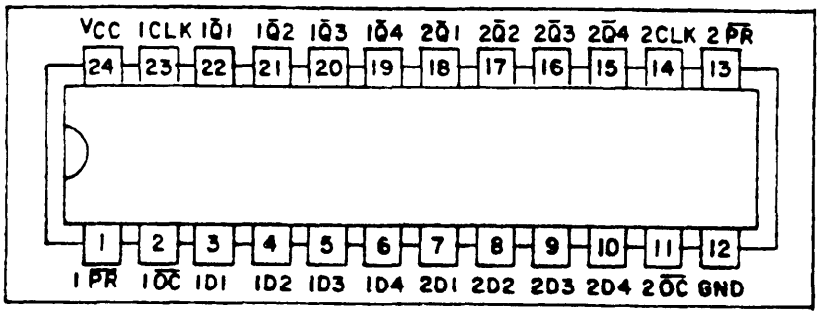


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 07

Cases L and K



Device type 07

Case 3

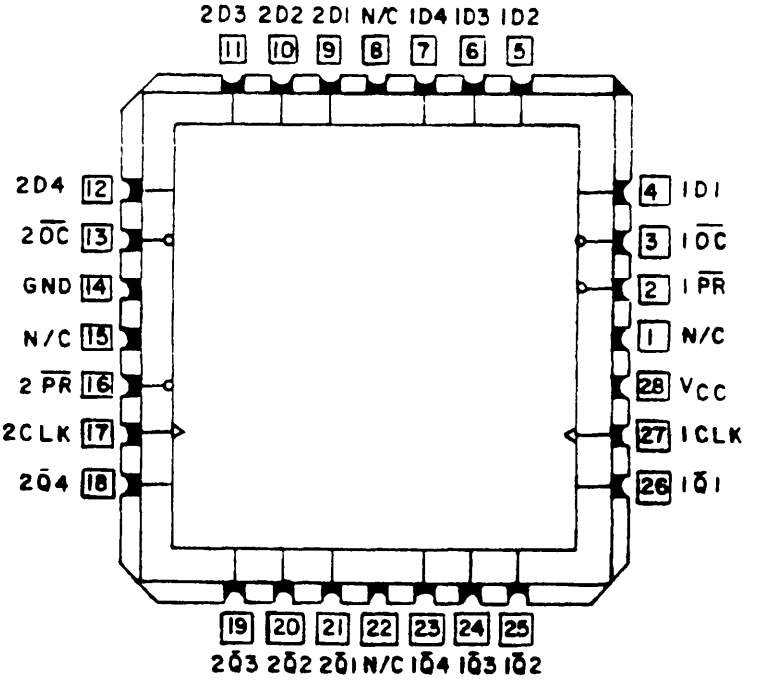
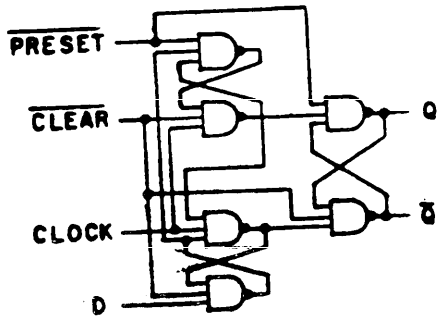


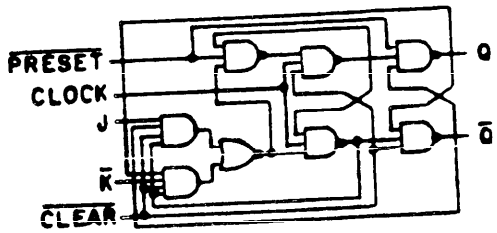
FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01



Device type 02



Device type 03

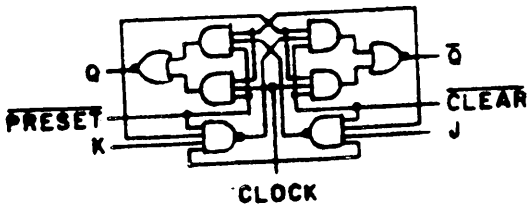
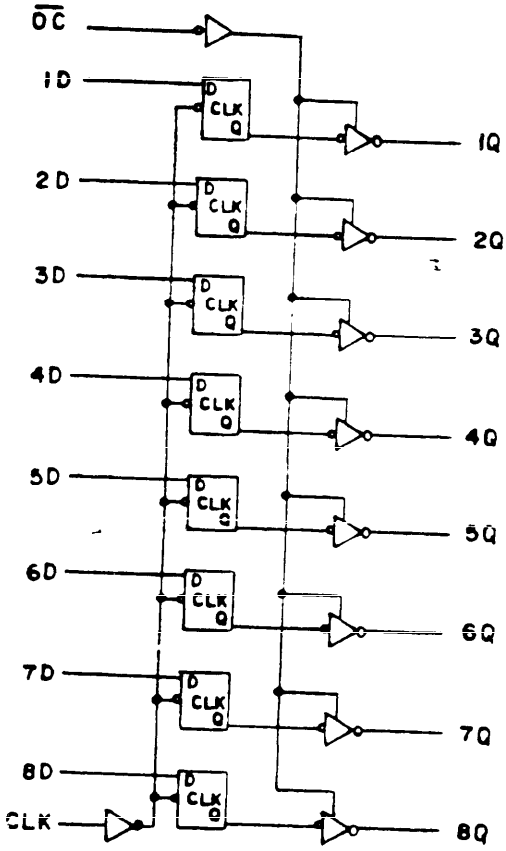


FIGURE 2. Logic diagrams.

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Device type 04



Device type 05

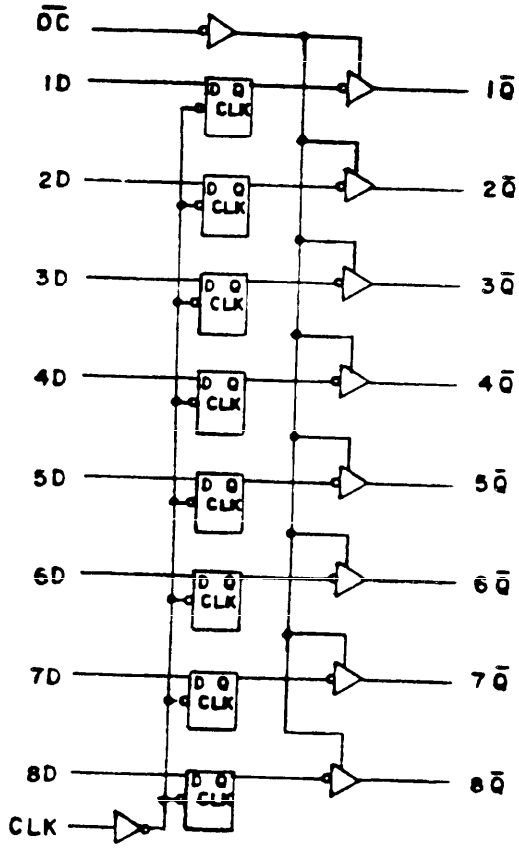


FIGURE 2. Logic diagrams - Continued.

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Device type 06

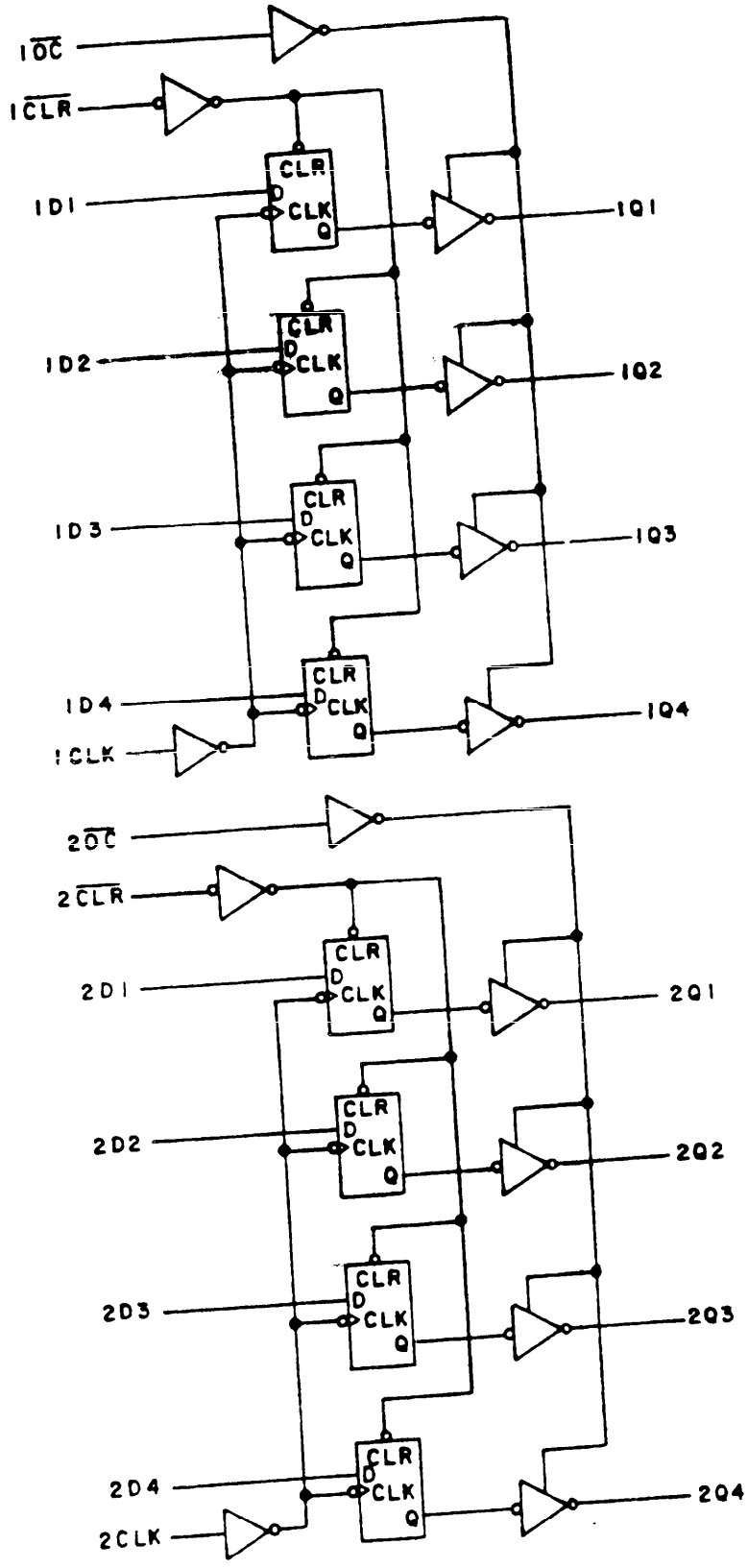


FIGURE 2. Logic diagrams - Continued.

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Device type 07

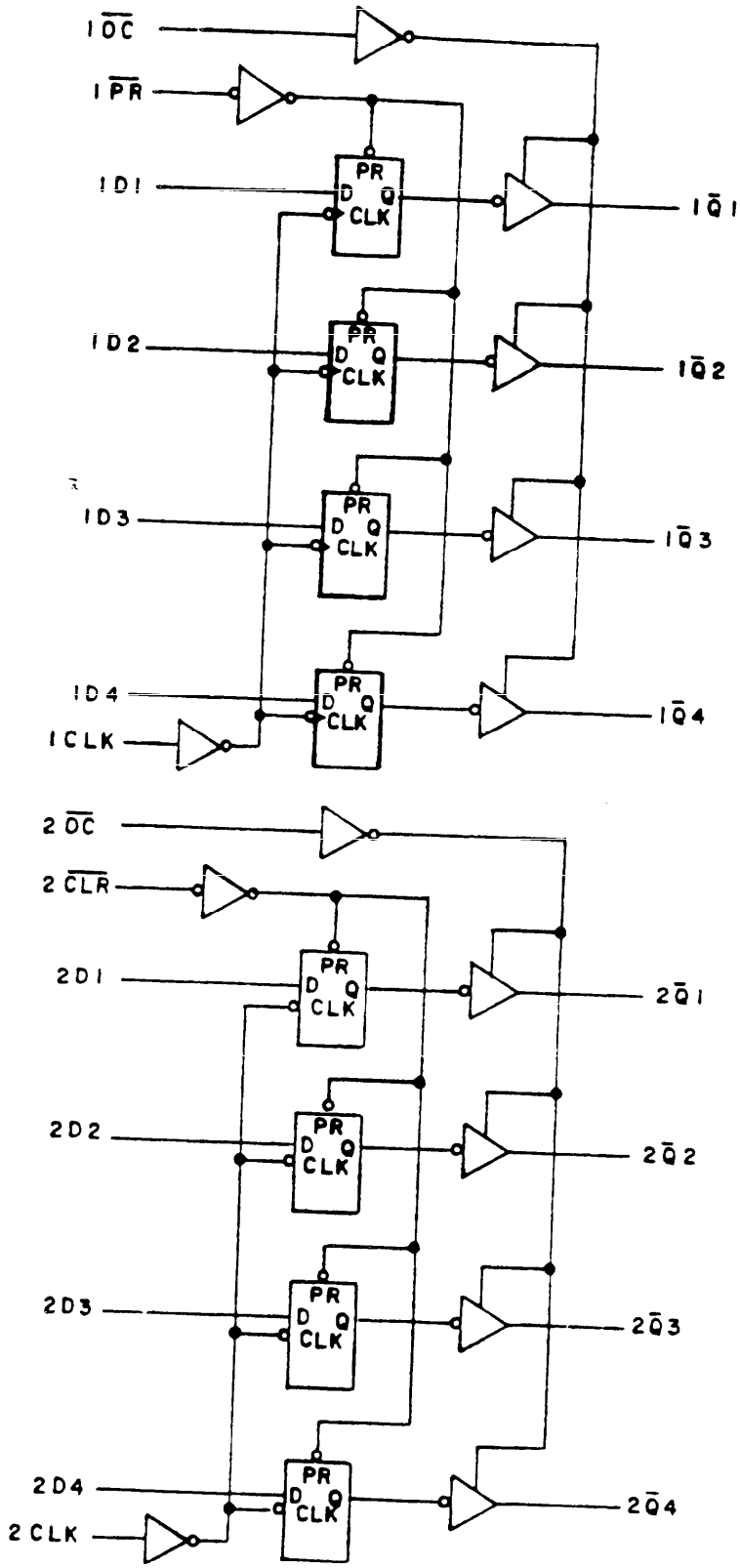


FIGURE 2. Logic diagrams - Continued.

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Device type 01

INPUTS				OUTPUTS	
$\overline{\text{PRESET}}$	$\overline{\text{CLEAR}}$	CLOCK	D	Q	$\overline{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H*	H*
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q_0	$\overline{Q}_0$

H = High level (steady state).
L = Low level (steady state).
X = Irrelevant
↑ = Transition from low to high level
Q_0 = The level of Q before the indicated steady state input conditions were established.
*This configuration is nonstable; that is, it will not persist when preset and clear inputs return to their inactive (high) level.

Device type 02

INPUTS					OUTPUTS	
$\overline{\text{PRESET}}$	$\overline{\text{CLEAR}}$	CLOCK	J	$\overline{K}$	Q	$\overline{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↑	L	L	L	H
H	H	↑	H	L	TOGGLE	
H	H	↑	L	H	Q_0	$\overline{Q}_0$
H	H	↑	H	H	H	L
H	H	L	X	X	Q_0	$\overline{Q}_0$

H = High level (steady state)
L = Low level (steady state)
X = Irrelevant
↑ = Transition from low to high level
Q_0 = The level of Q before the indicated steady state input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each ↑ clock transition.
*This configuration is nonstable; that is, it will not presist when preset and clear inputs return to their inactive (high) level.

FIGURE 3. Truth tables.

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Device type 03

INPUTS					OUTPUTS	
$\overline{PRESET}$	$\overline{CLEAR}$	CLOCK	J	K	Q	$\overline{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	+	L	L	Q_0	$\overline{Q}_0$
H	H	+	L	H	L	H
H	H	+	H	L	H	L
H	H	+	H	H	TOGGLE	
H	H	H	X	X	Q_0	$\overline{Q}_0$

H = High level (steady state)
L = Low level (steady state)
X = Irrelevant
+ = Transition from high to low level
Q_0 = The level of Q before the indicated steady state input conditions were established.
TOGGLE: Each output changes to the complement of its previous level on each + clock transition.
*This configuration is nonstable; that is, it will not presist when preset and clear inputs return to their inactive (high) level.

Device type 04

OUTPUT CONTROL	CLOCK	D	OUTPUT Q
L	+	H	H
L	+	L	L
L	L	X	Q_0
H	X	X	Z

H = High level (steady state)
L = Low level (steady state)
X = Irrelevant
+ = Transition from low to high level
Q_0 = The level of Q before the indicated steady state input conditions were established.
Z = High Impedance state

FIGURE 3. Truth tables - Continued.

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Device type 05

OUTPUT CONTROL	CLOCK	D	OUTPUT $\overline{Q}$
L	↑	H	L
L	↑	L	H
L	L	X	$\overline{Q}_0$
H	X	X	Z

H = High level (steady state)
L = Low level (steady state)
X = Irrelevant
↑ = Transition from low to high level
$\overline{Q}_0$ = The level of $\overline{Q}$ before the indicated steady state input conditions were established.
Z = High impedance state

Device type 06

CLEAR	DATA	CLOCK	OUTPUT CONTROL	OUTPUT
$\overline{CLR}$	D	CLK	$\overline{OC}$	Q
X	X	X	H	Z
L	X	X	L	L
H	H	↑	L	H
H	L	↑	L	L
H	X	L	L	Q_0

H = High level (steady state)
L = Low level (steady state)
X = Irrelevant
↑ = Transition from low to high level
Q_0 = The level of Q before the indicated steady state input conditions were established.
Z = High impedance state

Device type 07

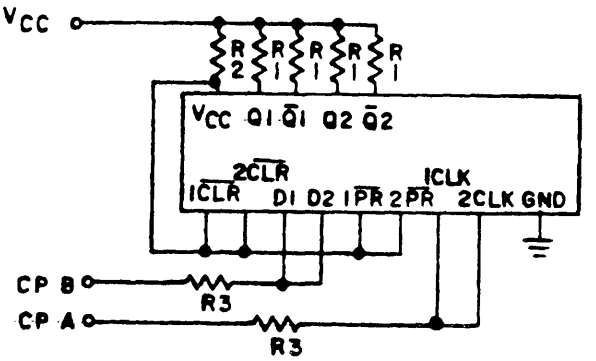
PRESET	DATA	CLOCK	OUTPUT CONTROL	OUTPUT
$\overline{PR}$	D	CLK	$\overline{OC}$	$\overline{Q}$
X	X	X	H	Z
L	X	X	L	L
H	H	↑	L	H
H	L	↑	L	$\overline{Q}_0$
H	X	L	L	

H = High level (steady state)
L = Low level (steady state)
X = Irrelevant
↑ = Transition from low to high level
$\overline{Q}_0$ = The level of $\overline{Q}$ before the indicated steady state input conditions were established.
Z = High impedance state

FIGURE 3. Truth tables - Continued.

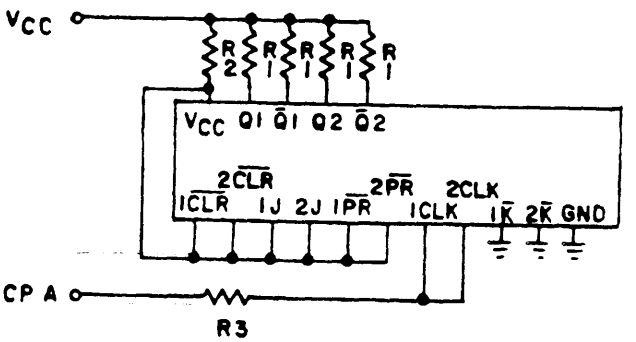
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:29Z
Check the source to verify that this is the current version before use.

Device type 01



NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device type 02

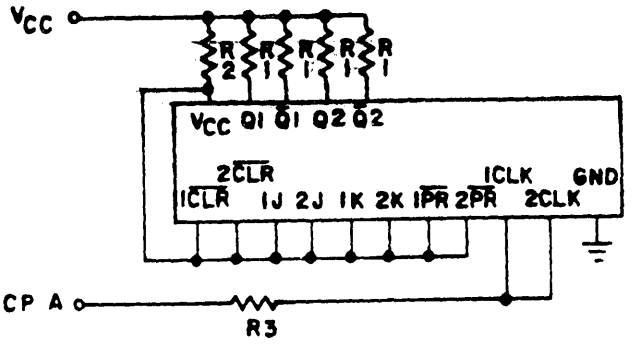


NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuits.

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Device type 03

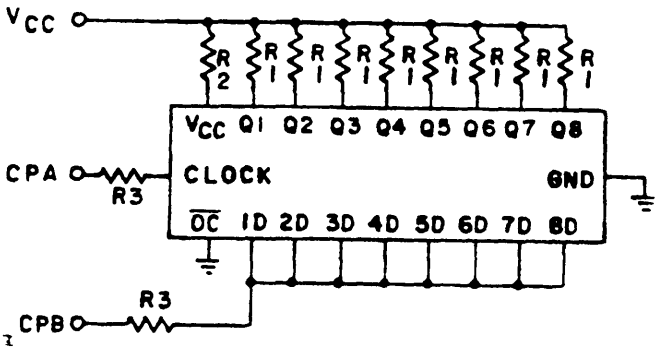


NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuits - Continued.

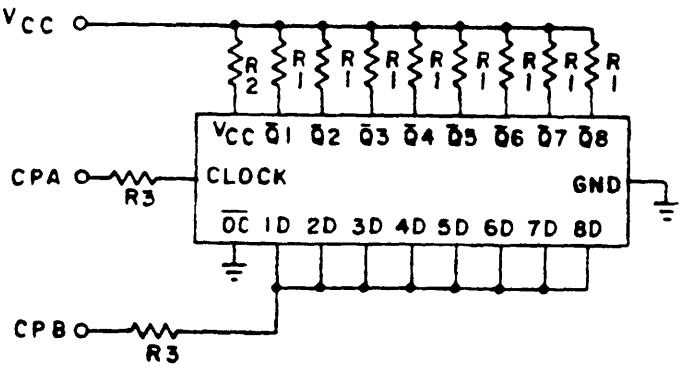
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Check the source to verify that this is the current version before use.

Device type 04



NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device type 05

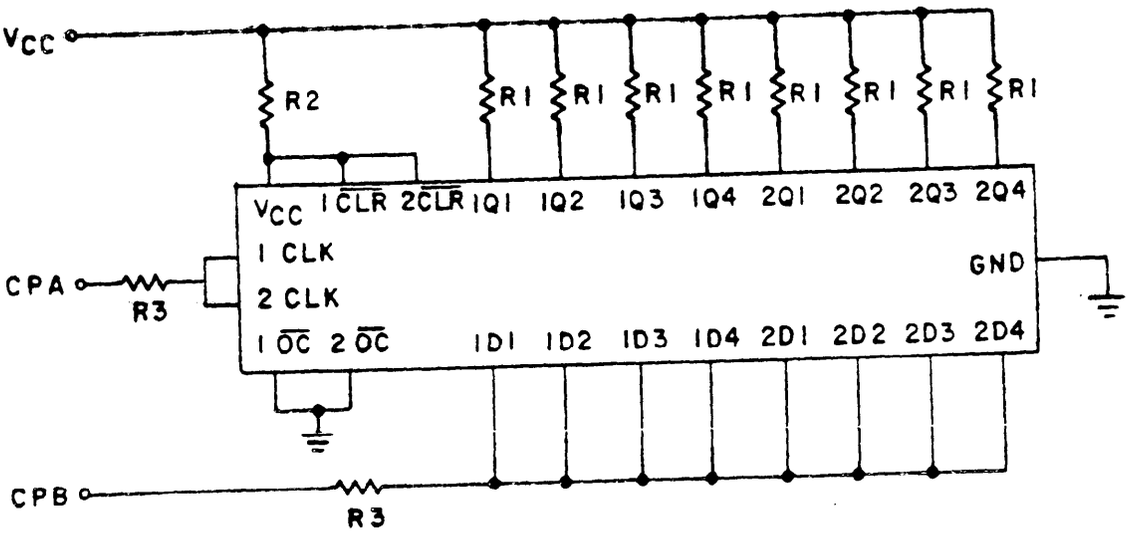


NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuits - Continued.

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Device type 06

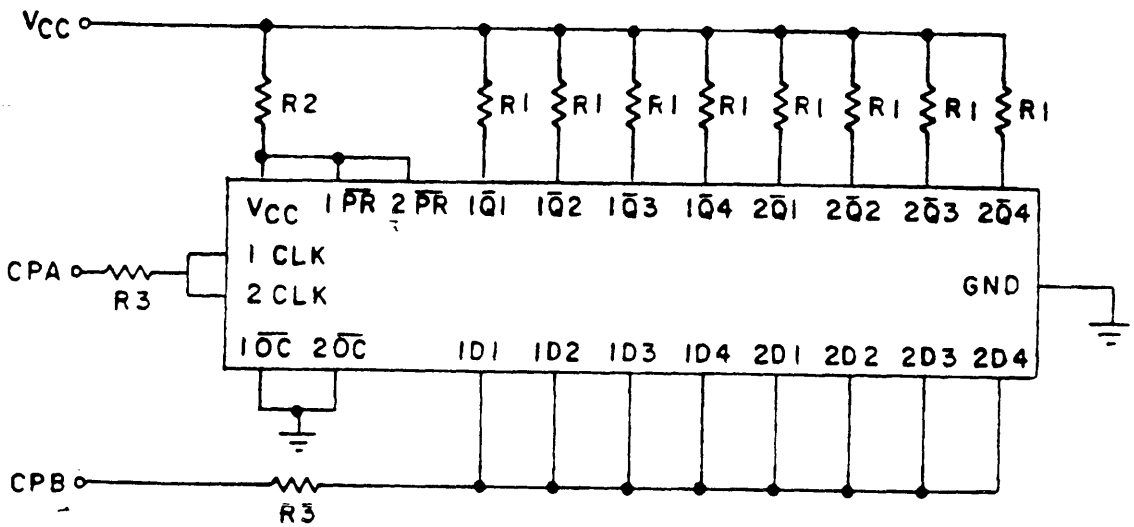


NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuits - Continued.

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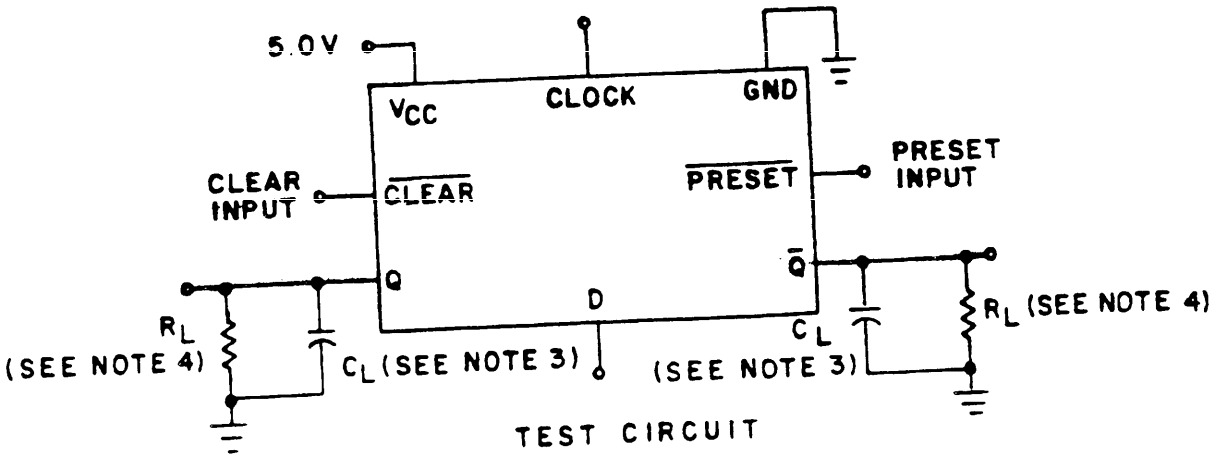
Device type 07



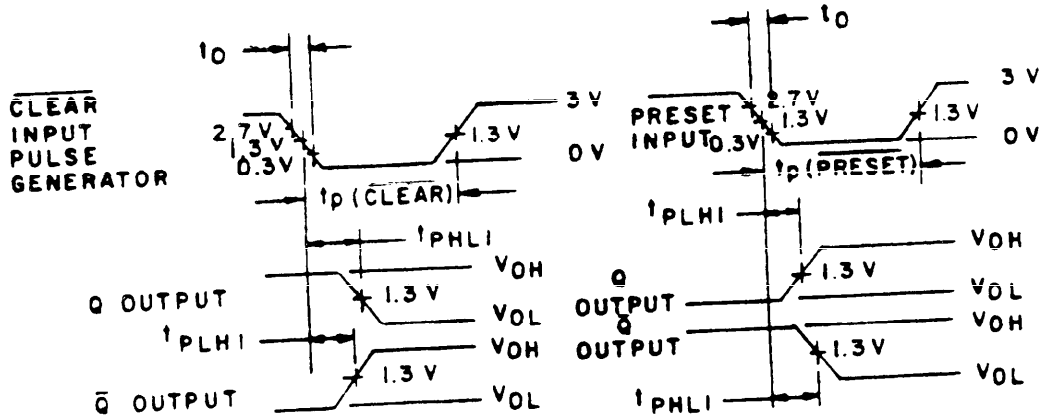
NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R = 1 kΩ ±5%; R3 = 27Ω ±5%.
3. V_{CC} and R2 shall be chosen to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuits - Continued.

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TEST CIRCUIT

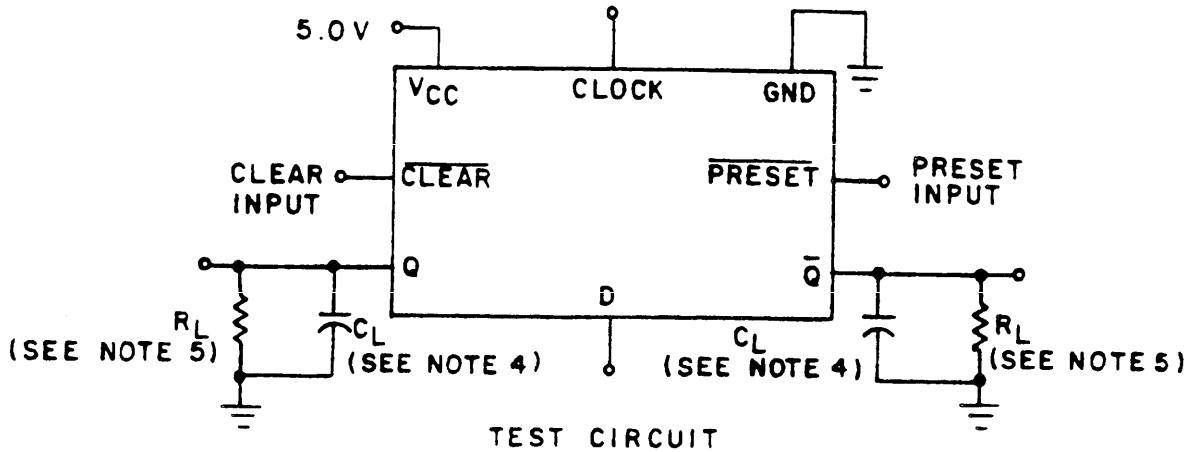


VOLTAGE WAVEFORMS

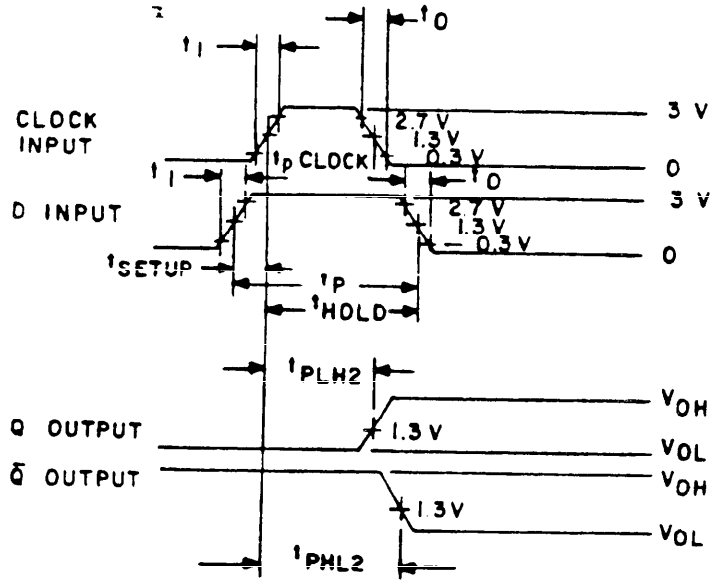
NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or D inputs.
2. Clear or preset input pulse characteristics: t_p (clear) = 15 ns; t_p (preset) = 15 ns; and PRR ≤ 1 MHz; t_0 = 6 ±1.5 ns.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
4. R_L = 499Ω ±1%.
5. When testing clear to output switching, preset input shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied (see table III).
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Clear and preset switching test circuit and waveforms (device type 01).

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TEST CIRCUIT



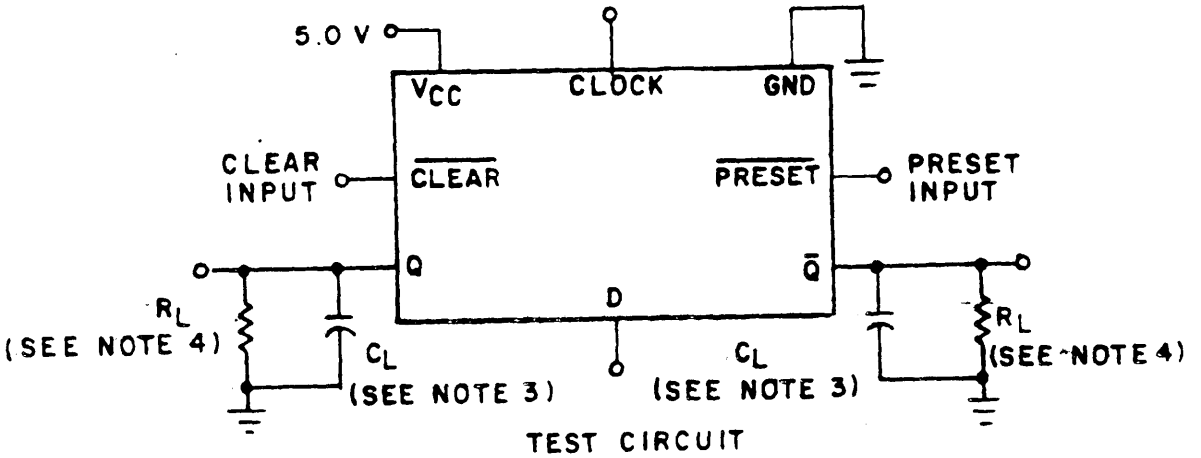
VOLTAGE WAVEFORMS

NOTES:

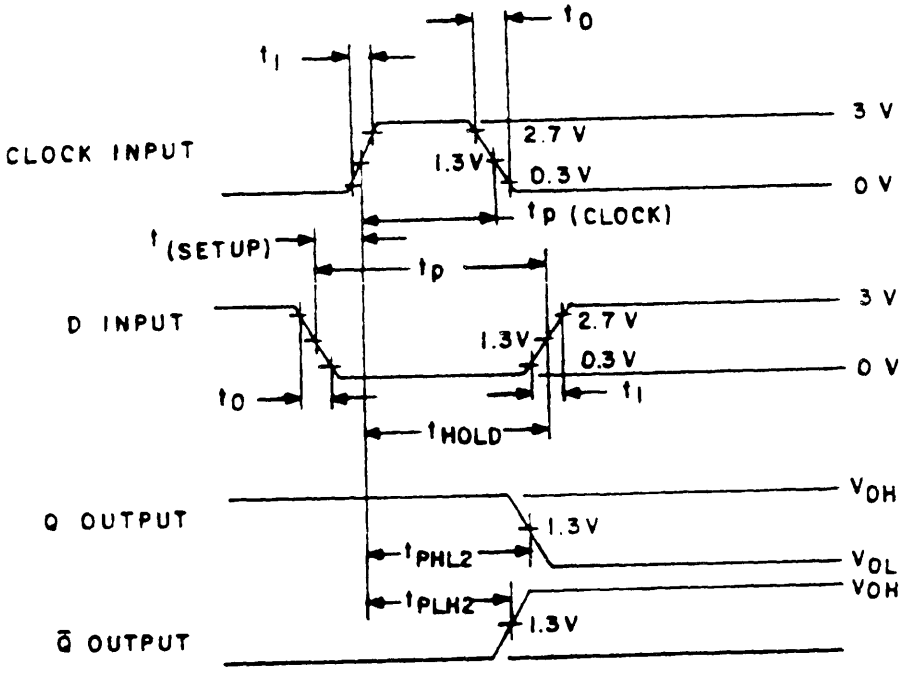
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{setup} = 15$ ns; $t_{hold} = 2$ ns; $t_p = 15$ ns; PRR is 50% on the clock PRR.
3. For f_{MAX}, the clock input pulse characteristics are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz.
   The D input pulse shall be one-half of the frequency of the clock and the D + and + shall be coincident with the clock +. $t_1 = t_0 = 6 \pm 1.5$ ns.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high-level data) device type 01.

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TEST CIRCUIT

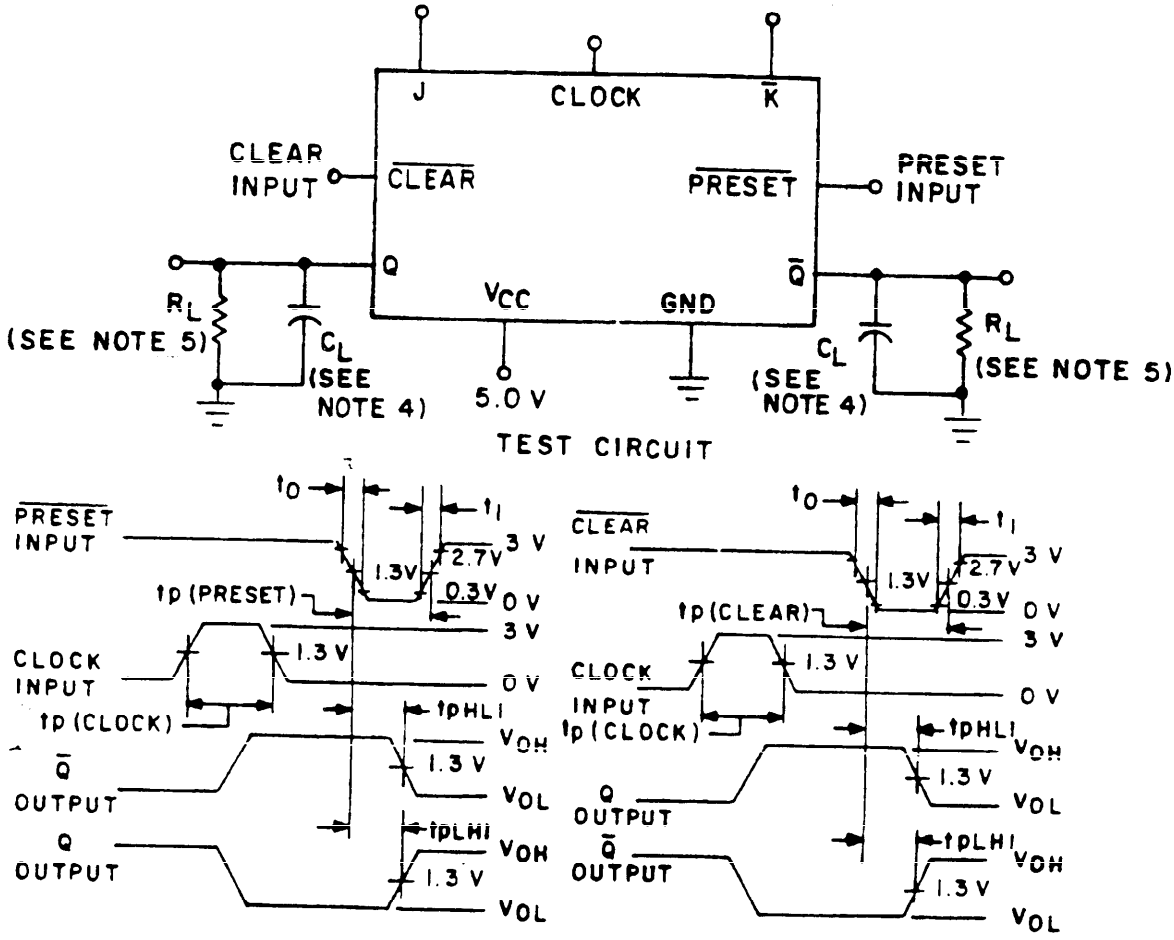


VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR = 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{setup} = 15$ ns; $t_{hold} = 2$ ns; $t_p = 15$ ns; PRR is 50% of the clock PRR.
3. $C_L = 50$ pF ±10% (including jig and probe capacitance without package in test fixture).
4. $R_L = 499\Omega \pm 1\%$.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Synchronous switching test circuit (low-level data) (device type 01).

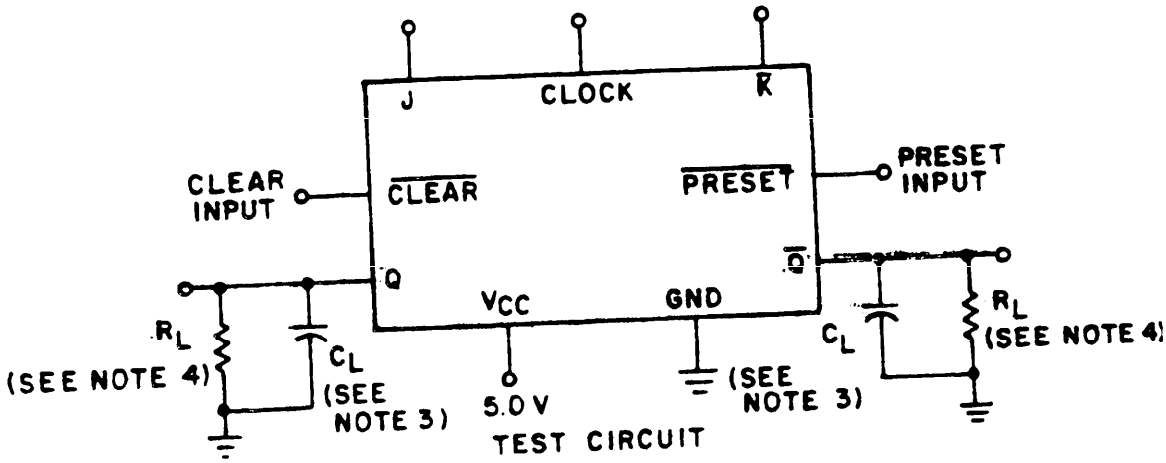
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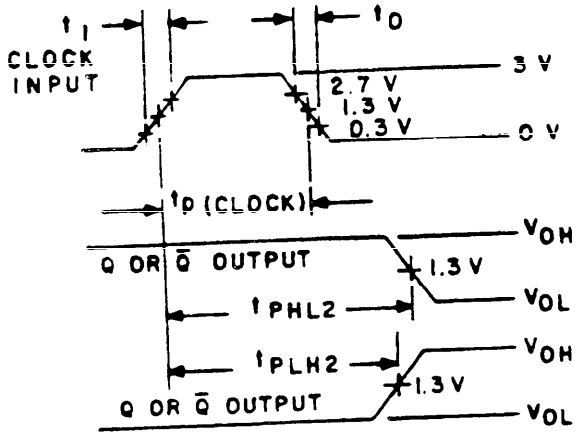
NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear and preset input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p ($\overline{\text{clear}}$) = 15 ns; t_p ($\overline{\text{preset}}$) = 15 ns; and PRR $\leq$ 1 MHz.
3. t_p (clock) = 16.5 ns min; PRR = same as clear and preset.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. When testing clear to output switching, preset input shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied (see table III).
7. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Clear and preset switching test circuit and waveforms (device type 02).

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TEST CIRCUIT

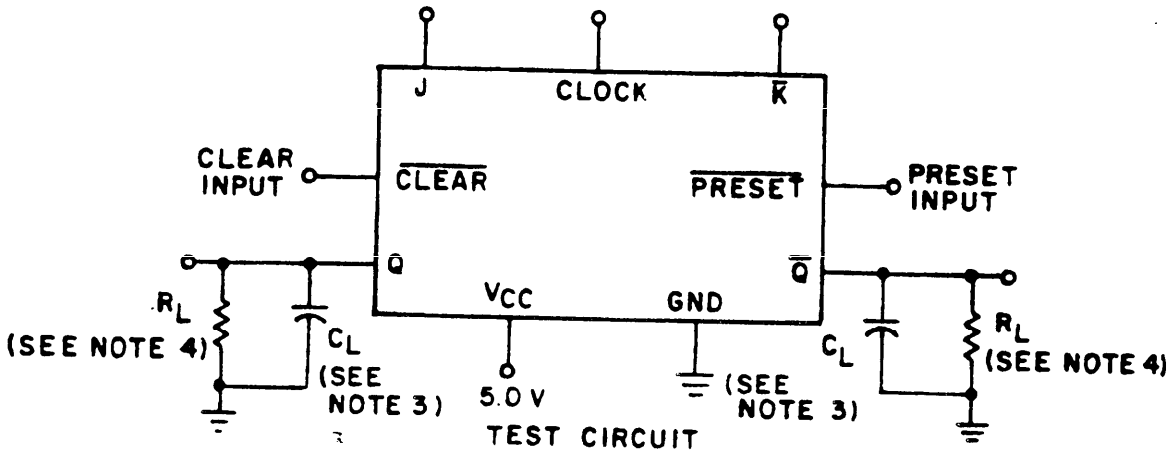


VOLTAGE WAVEFORMS

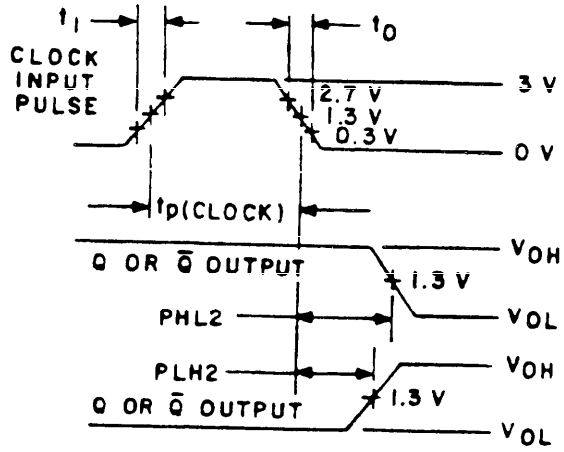
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characterisitcs are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
4. R_L = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (device type 02).

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TEST CIRCUIT

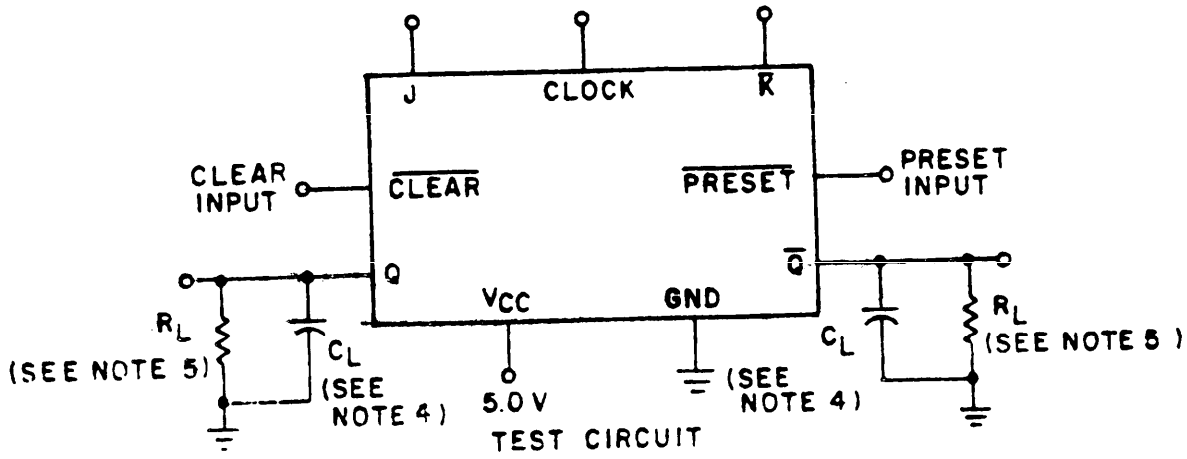


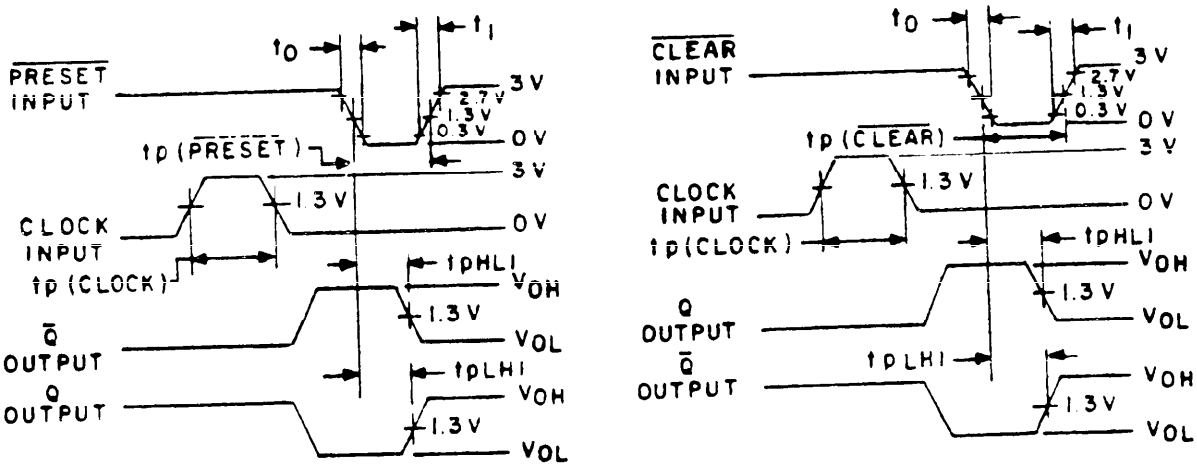
VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 \leq 6 \pm 1.5$ ns; t_p (clock) = 20 ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 20 ns; PRR = 25 MHz.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
4. R_L = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (device type 03).

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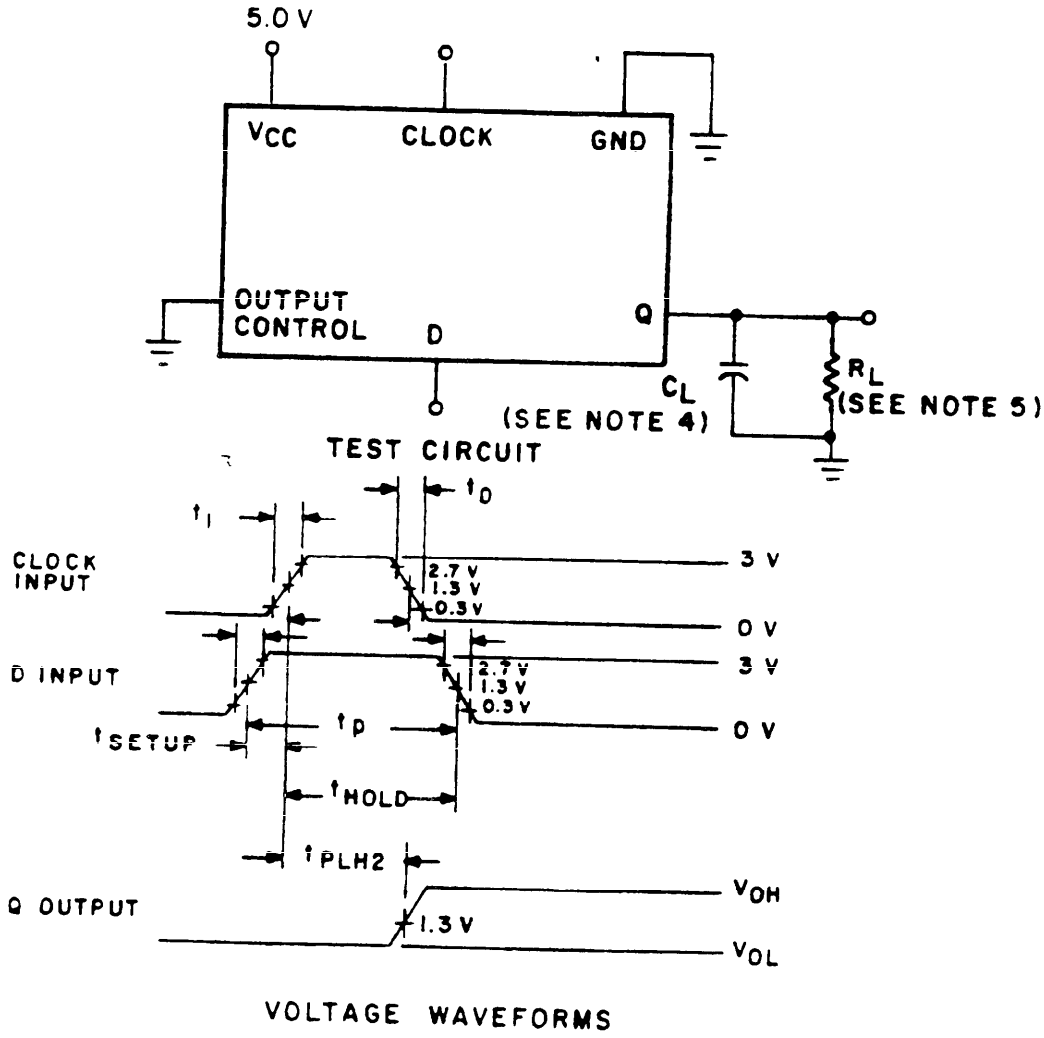


VOLTAGE WAVEFORMS

NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear and preset input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (Clear) = 15 ns; t_p (preset) = 15 ns; and PRR $\leq$ 1 Mhz.
3. t_p (clock) = 20 ns minimum; PRR = clear or preset PRR.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. When testing clear to output switching, present inputs shall have a logical "1" voltage applied. When testing preset to output switching, clear input shall have a logical "1" voltage applied (see table III).

FIGURE 5. Clear and preset switching test circuit and waveforms (device type 03).

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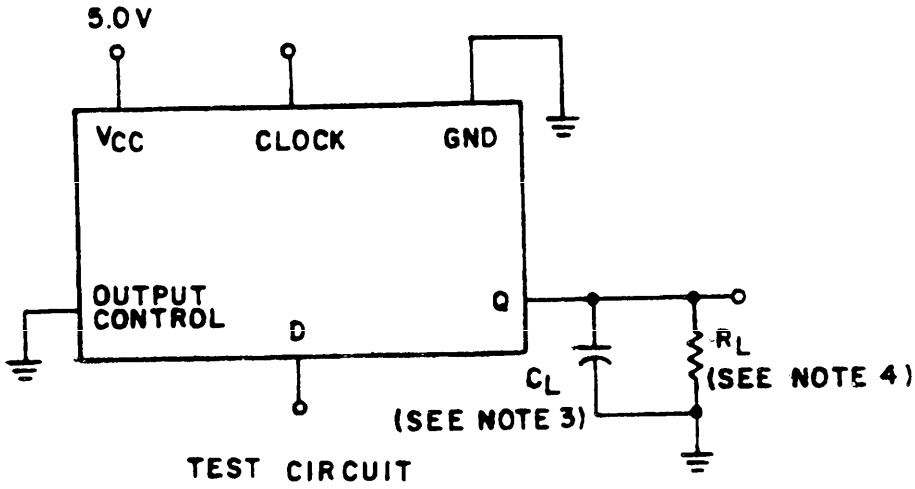


VOLTAGE WAVEFORMS

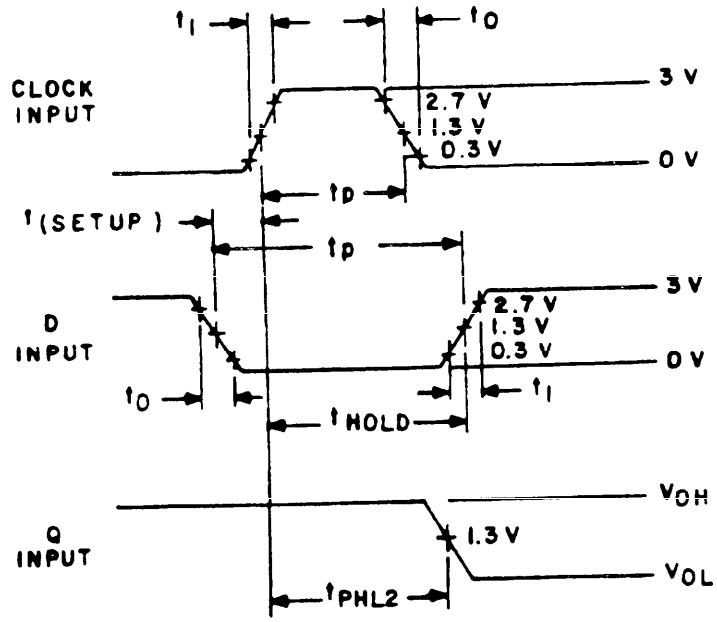
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 15 ns; t_{hold} = 4 ns; t_p = 19 ns; PRR is 50% of the clock PRR.
3. For f_{MAX}, the clock input pulse characteristics are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz.
   The D input pulse shall be one-half of the frequency of the clock and the D ↑ and ↓ shall be coincident with the clock ↓. $t_1 = t_0 = 6 \pm 1.5$ ns.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high-level data) (device type 04).

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TEST CIRCUIT

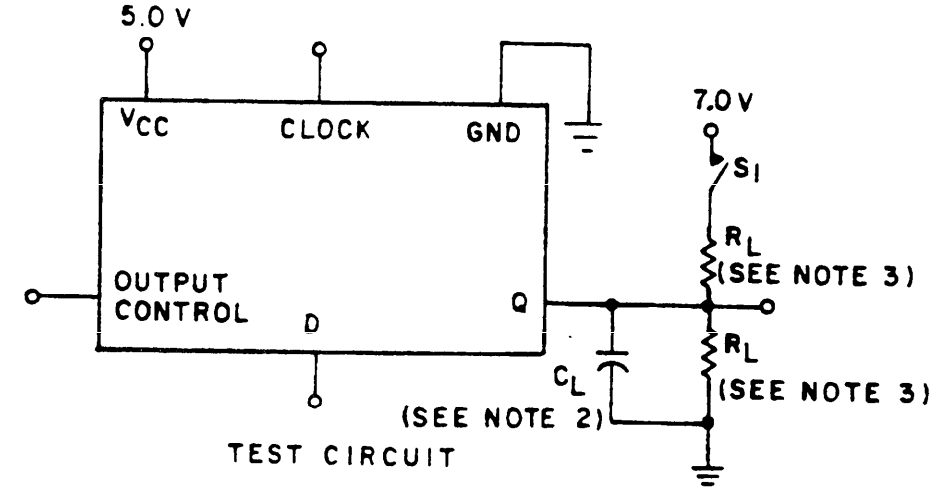


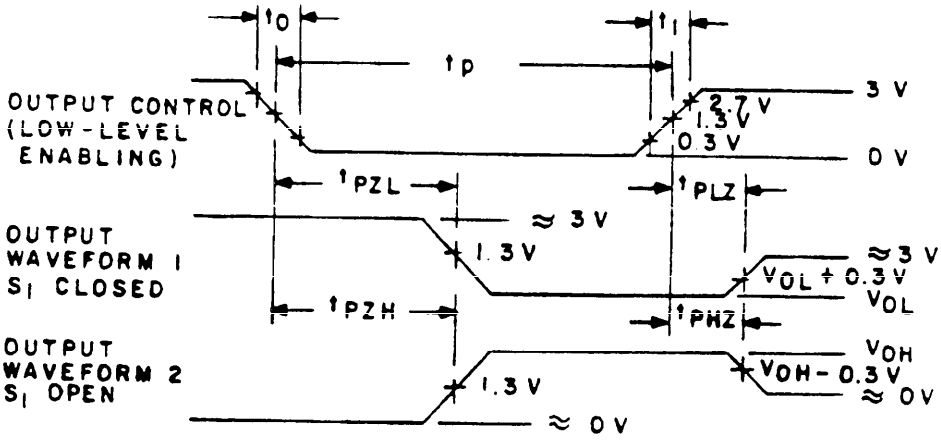
VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 15 ns; t_{hold} = 4 ns; t_p = 19 ns; PRR is 50% of the clock PRR.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
4. R_L = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (low-level data) (device type 04).

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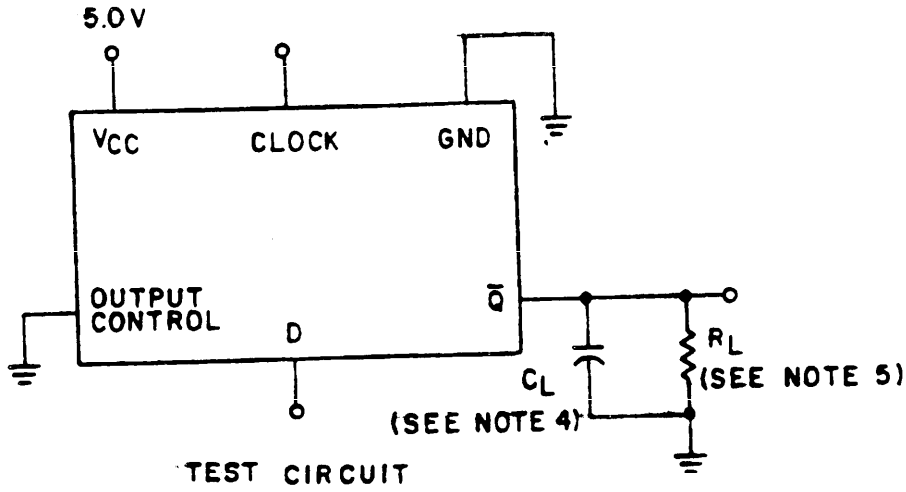


VOLTAGE WAVEFORMS

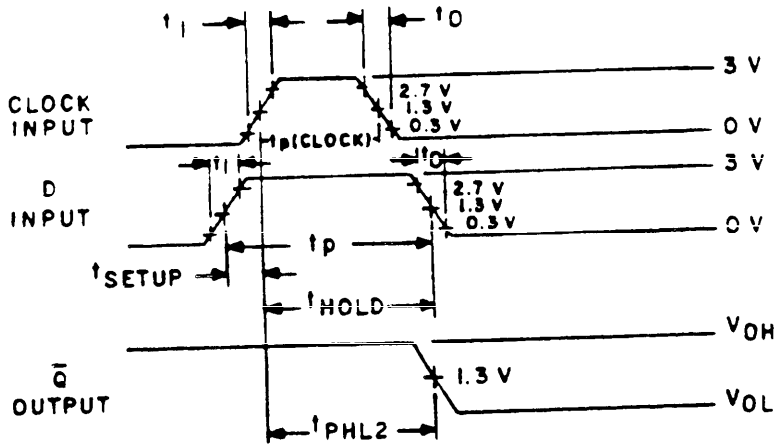
NOTES:
1. Output control input characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p \geq 200$ ns; PRR $\leq$ 1 MHz.
2. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
3. R_L = 499Ω ±5%.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Tri-state switching test circuit (device type 04).

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TEST CIRCUIT

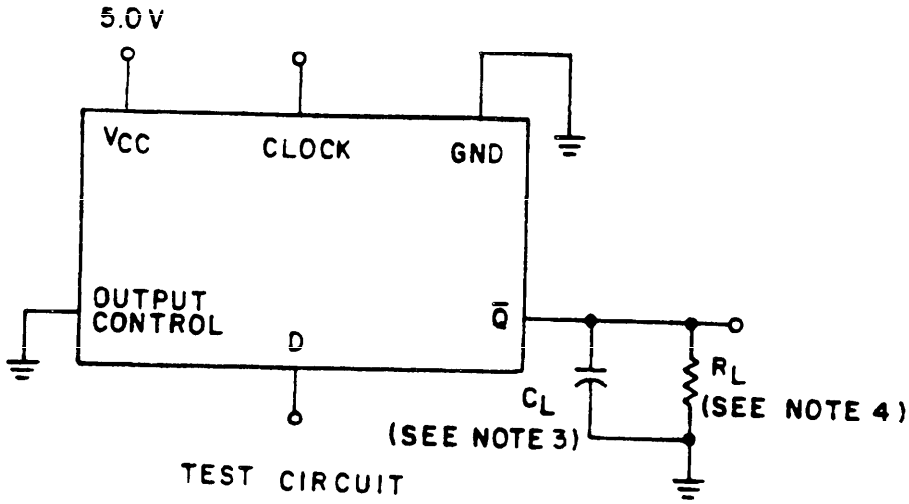


VOLTAGE WAVEFORMS

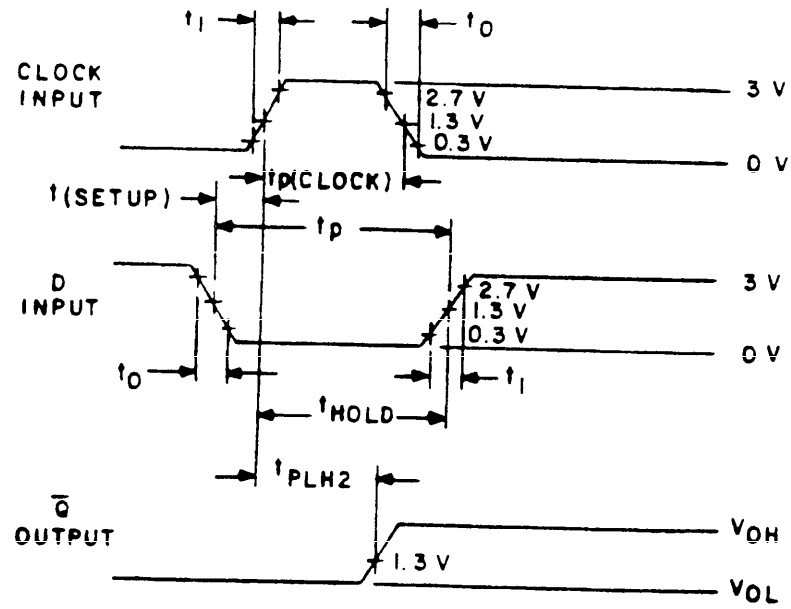
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 15 ns; t_{hold} = 4 ns; t_p = 19 ns; PRR is 50% of the clock PRR.
3. For f_{MAX}, the clock input pulse characteristics are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz.
   The D input pulse shall be one-half of the frequency of the clock and the D + and + shall be coincident with the clock +. $t_1 = t_0 = 6 \pm 1.5$ ns.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be amde with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high-level data) (device type 05).

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TEST CIRCUIT

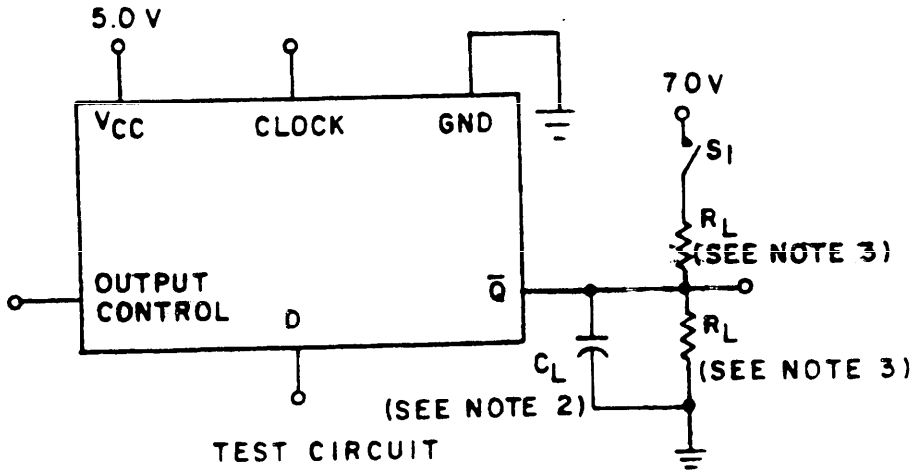


VOLTAGE WAVEFORMS

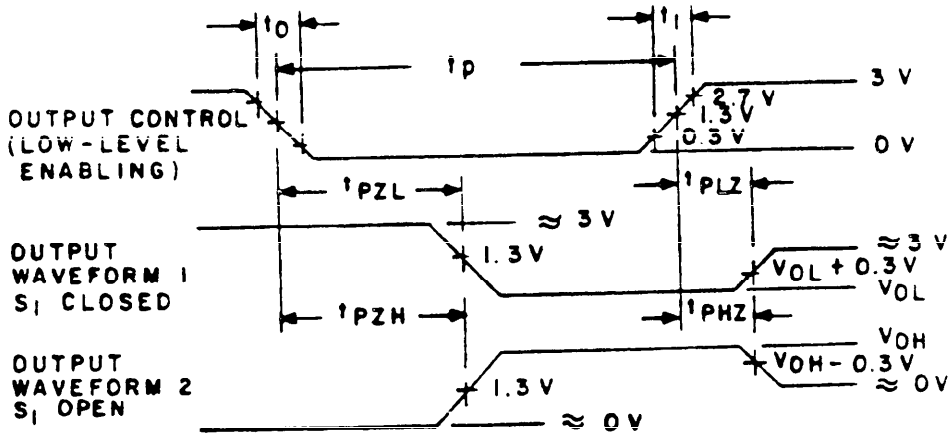
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 15 ns; t_{hold} = 4 ns; t_p = 19 ns; PRR is 50% of the clock PRR.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
4. R_L = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (low-level data) (device type 05).

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TEST CIRCUIT

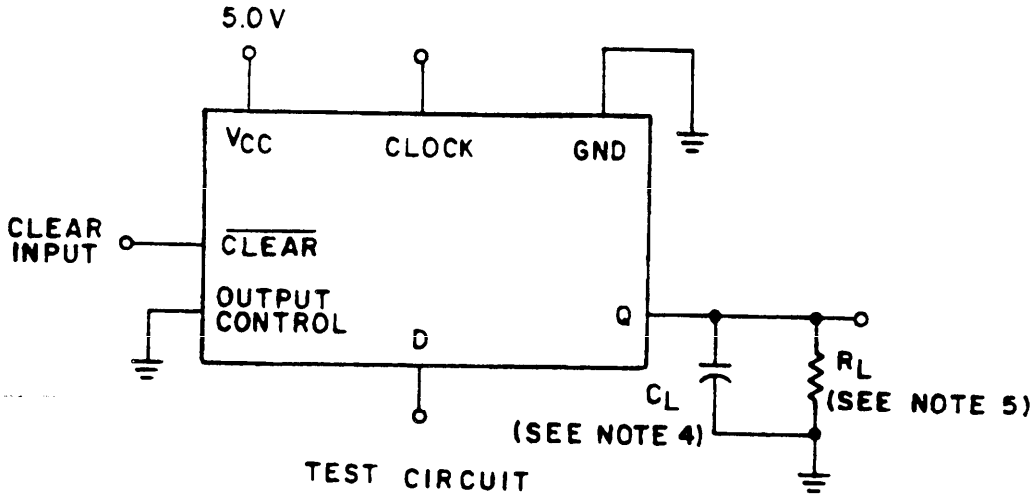


VOLTAGE WAVEFORMS

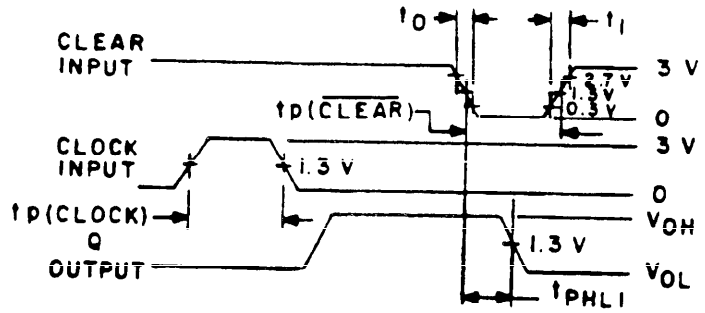
NOTES:
1. Output control input characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p \geq 200$ ns; PRR $\leq$ 1 MHz.
2. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
3. R_L = 499Ω ±5%.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Tri-state switching test circuit (device type 05).

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TEST CIRCUIT

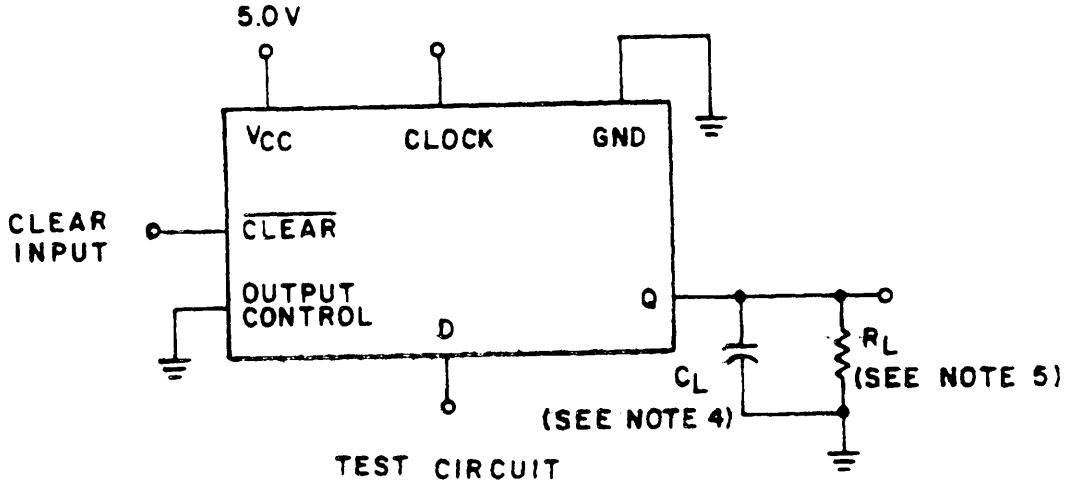


VOLTAGE WAVEFORMS

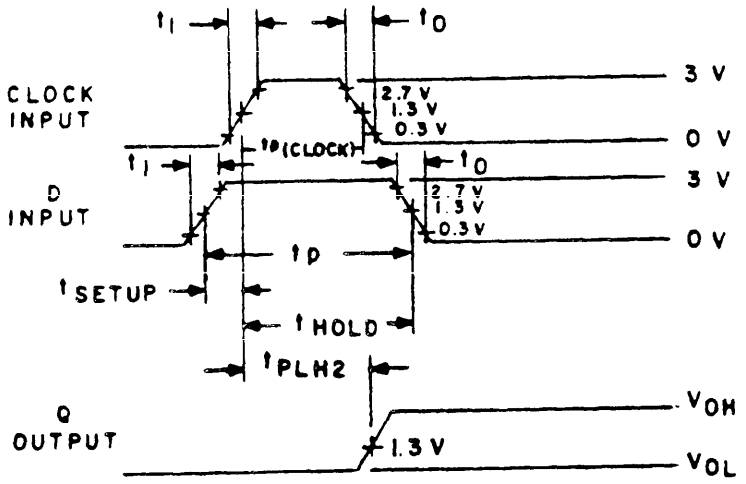
NOTES:
1. Clear inputs dominate regardless of the state of clock or D inputs.
2. Clear input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clear) = 10 ns; PRR $\leq$ 1 MHz.
3. t_p (clock) = 16.5 ns minimum; PRR = clear PRR.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Clear switching test circuit and waveforms (device type 06).

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TEST CIRCUIT

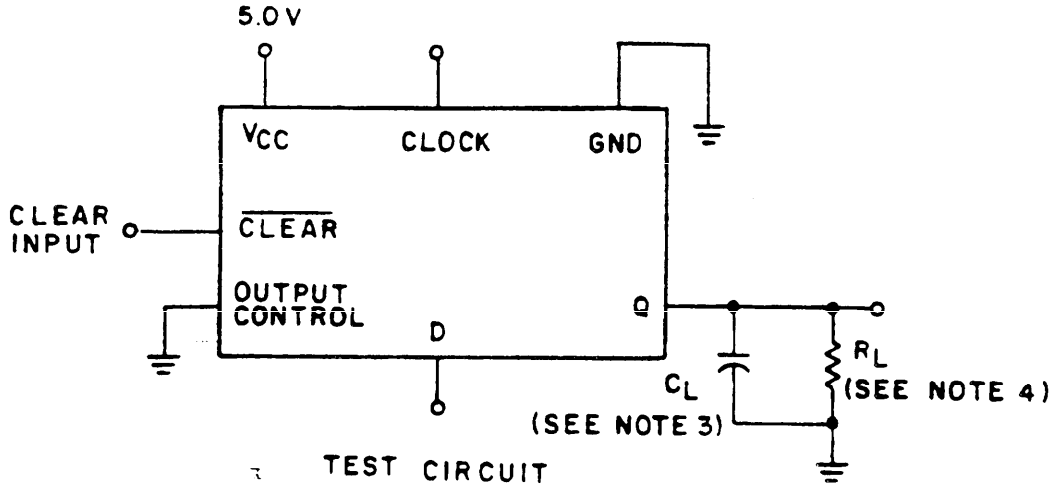


VOLTAGE WAVEFORMS

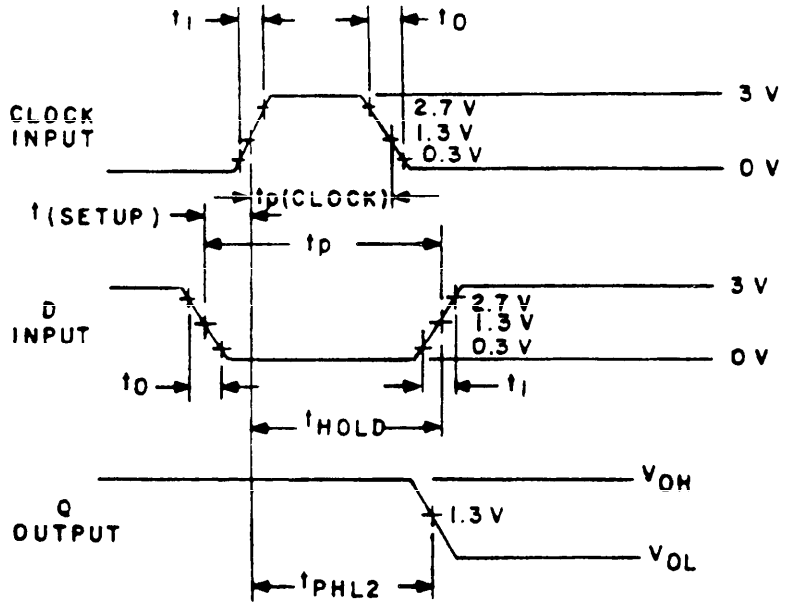
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{setup} = 15$ ns; $t_{hold} = 4$ ns; $t_p = 19$ ns; PRR is 50% of the clock PRR.
3. For f_{MAX}, the clock input pulse characteristics are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz.
   The D input pulse shall be one-half of the frequency of the clock and the D + and + shall be coincident with the clock . $t_1 = t_0 = 6 \pm 1.5$ ns.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high-level data) (device type 01).

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TEST CIRCUIT

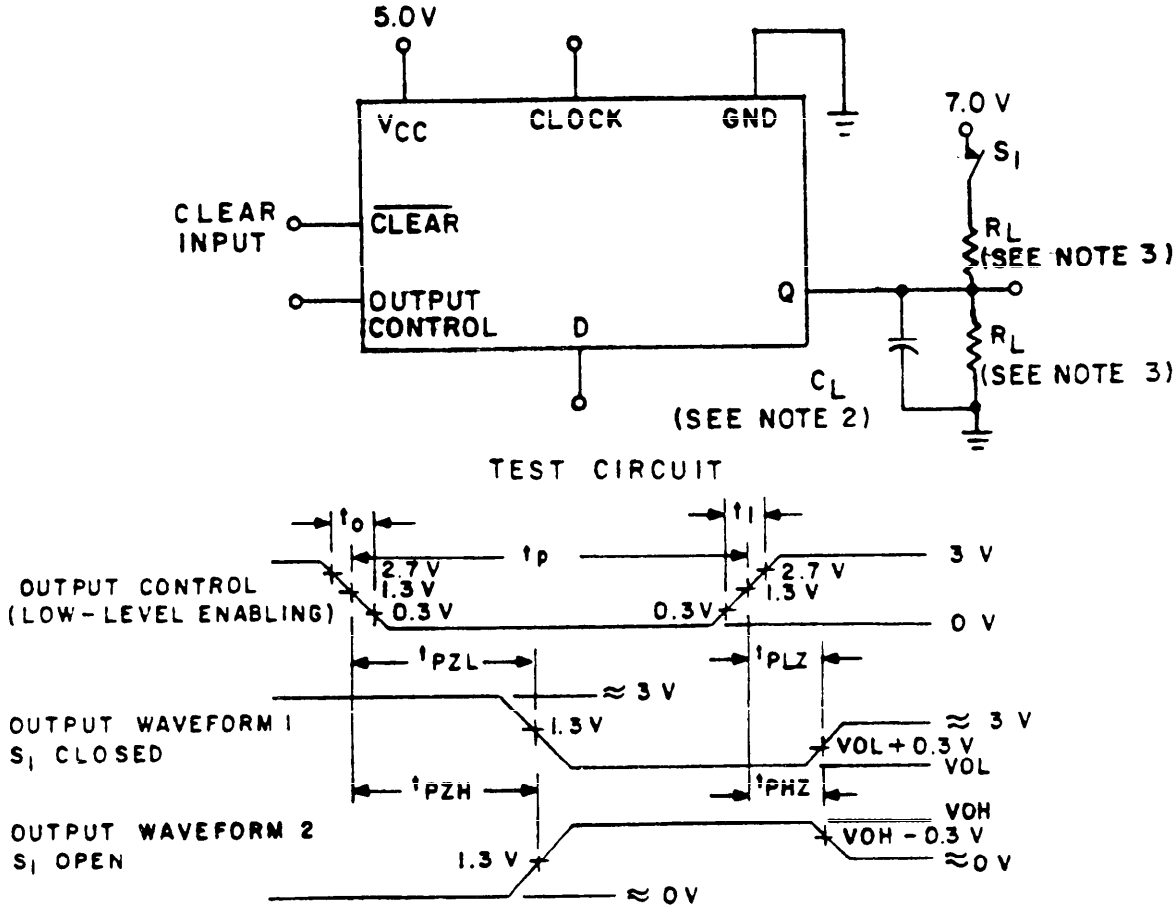


VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 15 ns; t_{hold} = 4 ns; t_p = 19 ns; PRR is 50% of the clock PRR.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package-in test fixture).
4. R_L = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (low-level data) (device type 06).

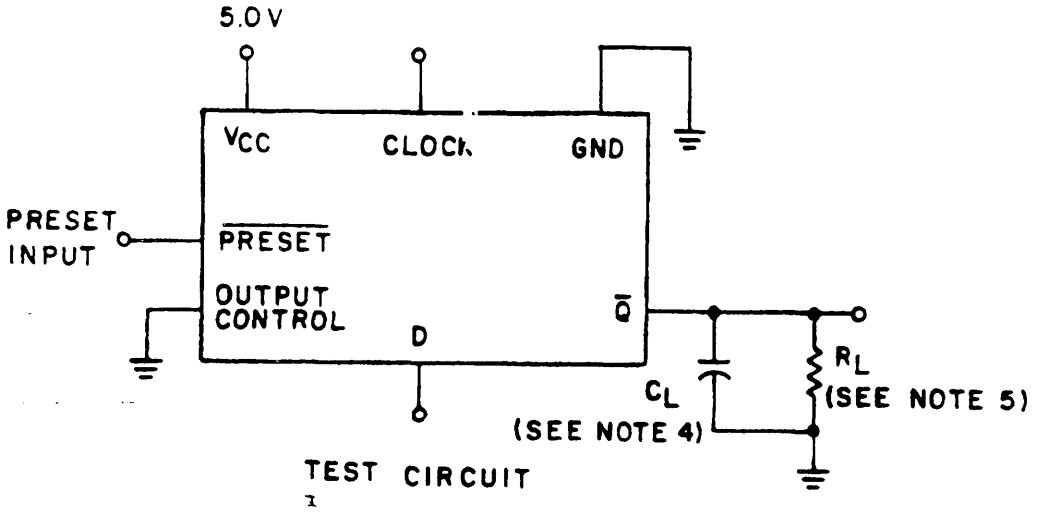
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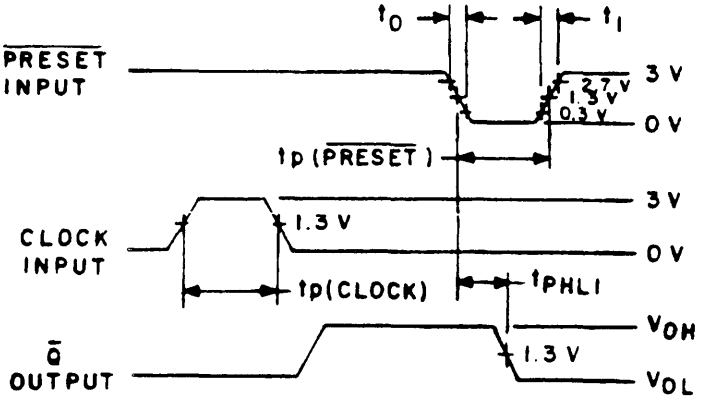
NOTES:
1. Output control input characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p \geq 200$ ns; PRF $\leq$ 1 MHz.
2. $C_L = 50$ pF ±10% (including jig and probe capacitance without package in test fixture).
3. $R_L = R_L = 499\Omega \pm 5\%$.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Tri-state switching test circuit (device type 06).

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TEST CIRCUIT

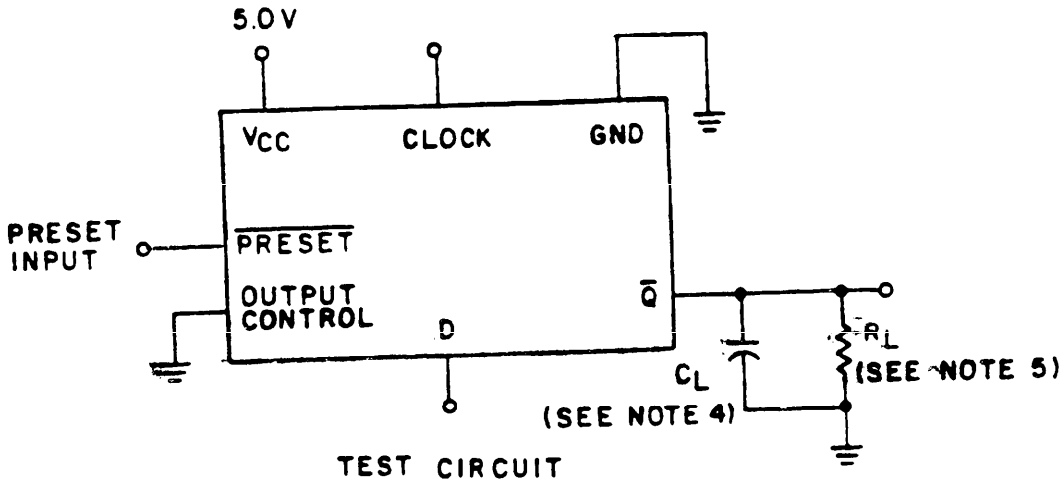


VOLTAGE WAVEFORMS

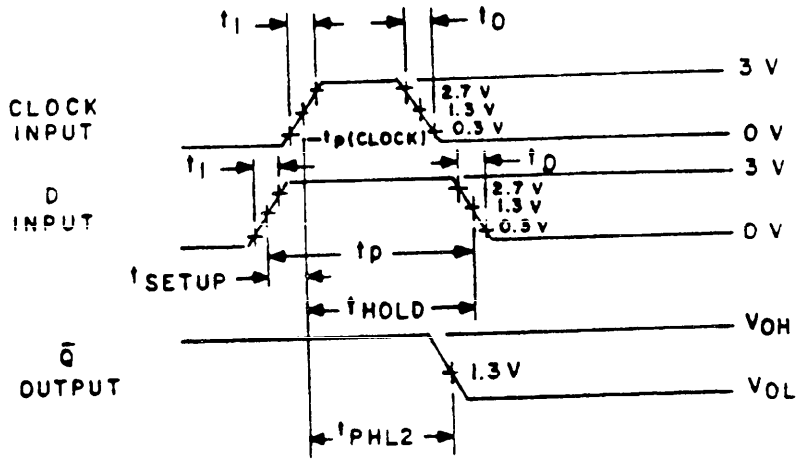
NOTES:
1. Preset inputs dominate regardless of the state of clock or D inputs.
2. Preset input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (preset) = 10 ns; PRR $\leq$ 1 MHz.
3. t_p (clock) = 16.5 ns min; PRR = Preset PRR.
4. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
5. R_L = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Preset switching test circuit and waveforms (device type 07).

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TEST CIRCUIT

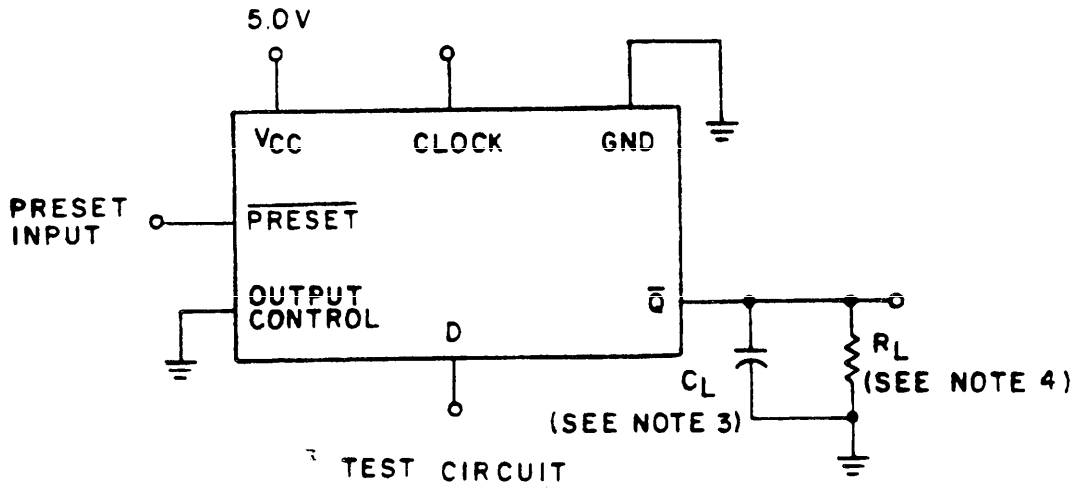


VOLTAGE WAVEFORMS

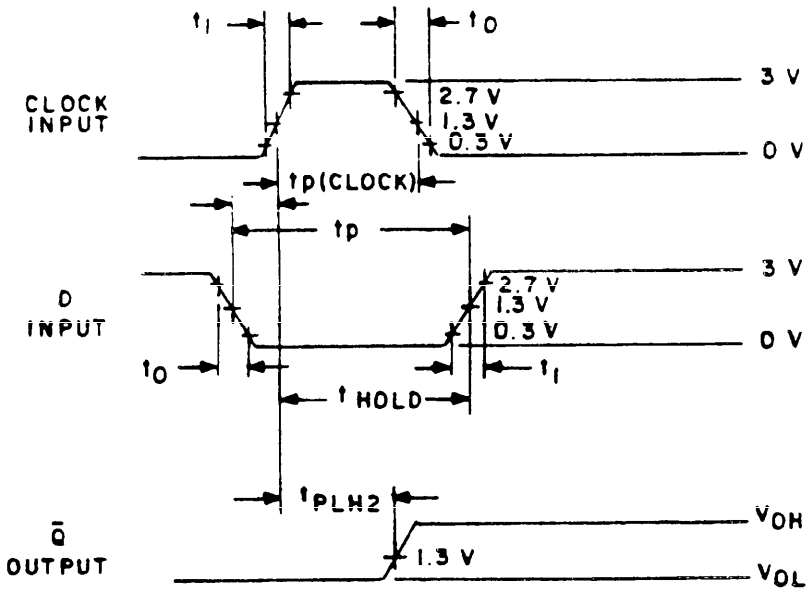
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{setup} = 15$ ns; $t_{hold} = 4$ ns; $t_p = 19$ ns; PRR is 50 of the clock PRR.
3. For f_{MAX}, the clock input pulse characteristics are as follows:
   $t_1 = t_0 \leq 3$ ns; t_p (clock) = 16.5 ns; PRR = 30 MHz.
   The D input pulse shall be one-half of the frequency of the clock and the D + and + shall be coincident with the clock +. $t_1 = t_0 = 6 \pm 1.5$ ns.
4. $C_L = 50$ pF ±10% (including jig and probe capacitance without package in test fixture).
5. $R_L = 499\Omega \pm 1\%$.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high-level data) (device type 07).

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TEST CIRCUIT

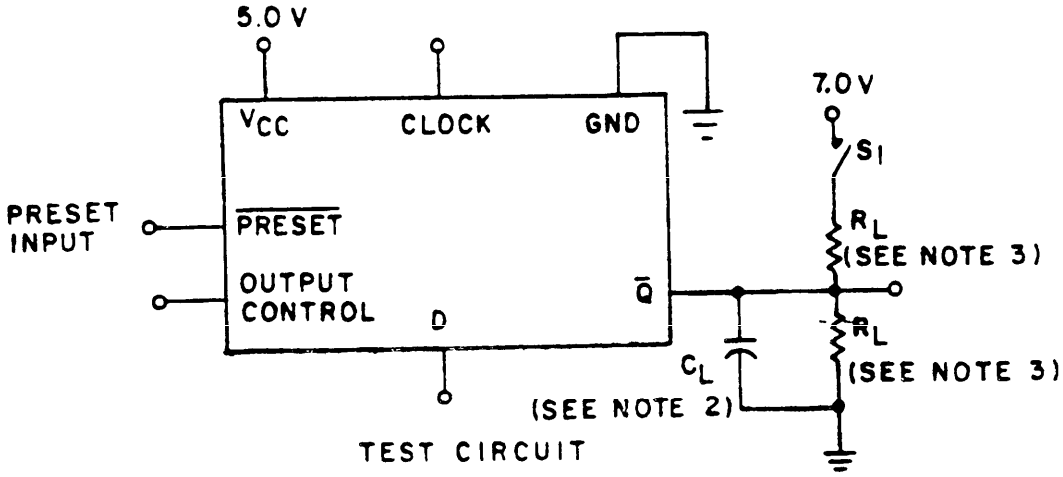


VOLTAGE WAVEFORMS

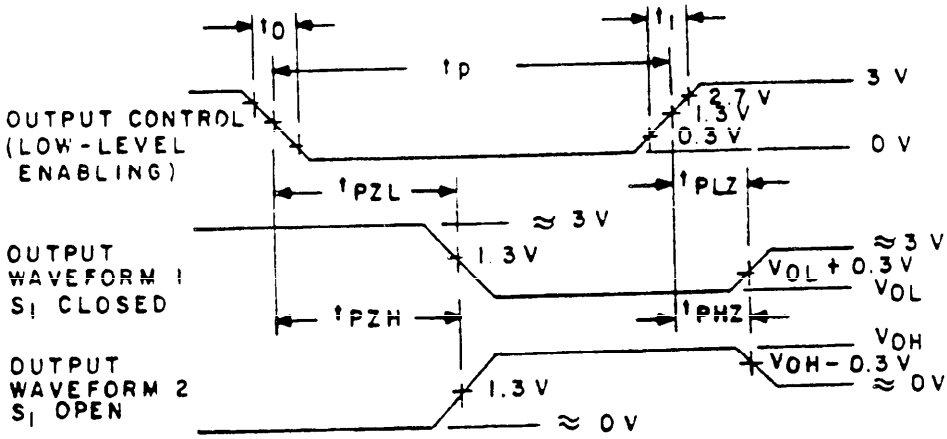
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_p (clock) = 16.5 ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; t_{setup} = 15 ns; t_{hold} = 4 ns; t_p = 19 ns; PRR is 50% of the clock PRR.
3. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
4. R_L = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (low-level data) (device type 07).

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Output control input characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p \geq 200$ ns; PRR $\leq$ 1 MHz.
2. C_L = 50 pF ±10% (including jig and probe capacitance without package in test fixture).
3. R_L = 499Ω ±5%.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Tri-state switching test circuit (device type 07).

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.4 V, or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases A, B, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1CLR	1D	1CLK	1PR	1Q	1Q	GND	2Q	2Q	2PR	2CLK	2D	2CLR	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	2.0 V	2.0 V	GND	0.8 V	-0.4 mA		GND			2.0 V	GND	2.0 V	0.8 V	4.5 V	1Q	2.5		V
			2	0.8 V	"	GND	2.0 V		-0.4 mA				0.8 V	GND	2.0 V	2.0 V		1Q			
			3	2.0 V	"	2/	"	-0.4 mA					2.0 V	2/	0.8 V	"		1Q			
			4	2.0 V	0.8 V	2/			-0.4 mA				2.0 V	2/	2.0 V			1Q			
			5								-0.4 mA	-0.4 mA	2.0 V	GND	2.0 V	0.8 V		2Q			
			6										0.8 V	GND	2.0 V	2.0 V		2Q			
			7								-0.4 mA	-0.4 mA	2.0 V	2/	0.8 V	"		2Q			
			8										2.0 V	2/	2.0 V	"		2Q			
	V_{OL}	3007	9	2.0 V	0.8 V	2/	2.0 V	4 mA										1Q		0.4	
			10	"	2.0 V	2/	2.0 V		4 mA									1Q			
			11	"	"	GND	0.8 V		4 mA									1Q			
			12	0.8 V	"	GND	2.0 V	4 mA										1Q			
			13									4 mA	2.0 V	2/	0.8 V	2.0 V		2Q			
			14								4 mA		2.0 V	2/	2.0 V	"		2Q			
			15								4 mA		0.8 V	GND	"	"		2Q			
			16									4 mA	2.0 V	GND	"	0.8 V		2Q			
	V_{IC}		17	-18 mA														1CLR		-1.5	
			18		-18 mA													1D		"	
			19			-18 mA												1CLK		"	
			20				-18 mA											1PR		"	
			21										-18 mA					2PR		"	
			22											-18 mA				2CLK		"	
			23												-18 mA			2D		"	
			24													-18 mA		2CLR		"	
	I_{IL1}	3009	25	5.0 V	0.4 V	5.0 V	GND										5.5 V	1D	8/	8/	µA
			26	5.0 V	GND	0.4 V	GND										"	1CLK	"	"	
			27										GND	0.4 V	GND	5.0 V	"	2CLK	"	"	
			28										GND	5.0 V	0.4 V	5.0 V	"	2D	"	"	
	I_{IL4}		29	0.4 V	5.0 V	5.0 V	GND											1CLR			
			30	GND	GND	GND	0.4 V											1PR			
			31										0.4 V	GND	GND	GND		2PR			
			32										GND	5.0 V	5.0 V	0.4 V		2CLR			
	I_{IH1}		33	GND	2.7 V	GND	GND											1D		20	
			34	GND	GND	2.7 V	GND											1CLK		"	
			35										GND	2.7 V	GND	GND		2CLK		"	
			36										GND	GND	2.7 V	GND		2D		"	
	I_{IH4}		37	2.7 V	GND	3/	GND											1CLR		40	
			38	GND	GND	3/	2.7 V											1PR		"	
			39										2.7 V	3/	GND	GND		2PR		"	
			40										GND	3/	GND	2.7 V		2CLR		"	
	I_{IH6}		41	GND	7.0 V	GND	GND											1D		100	
			42	GND	GND	7.0 V	GND											1CLK		"	
			43										GND	7.0 V	GND	GND		2CLK		"	
			44										GND	GND	7.0 V	GND		2D		"	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be High ≥2.0 V, or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 1/ 2	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Cases A, B, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19		Min	Max	
			Test no.	1CLR	1D	1CLK	1PR	1Q	1Q	GND	2Q	2Q	2PR	2CLK	2D	2CLR	V_{CC}									
1 T_C = 25°C	I_{IH9}	3009	45	7.0 V	GND	GND	J/	GND		GND							5.0 V	1CLR						200	μA
			46	GND	GND	J/	7.0 V			-			7.0 V	3/	GND	GND	-	1PR						-	-
			47							-			GND	3/	GND	7.0 V	-	2PR						-	-
			48							-							-	2CLR						-	-
	I_Q 4/	3011	49	5.0 V			GND	2.25 V	2.25 V	-	2.25 V						-	1Q					-30	-112	mA
			50	GND			5.0 V			-							-	1Q					-	-	-
			51							-		2.25 V					-	2Q					-	-	-
			52							-			5.0 V			GND	-	2Q					-	-	-
													GND			5.0 V									
	I_{CC}	3005	53	5.5 V	GND	GND	GND	GND		-							-	V_{CC}						40	-
			54	GND				5.5 V		-			GND	GND	GND	5.5 V	-	V_{CC}						-	-
													GND	GND	GND	GND									
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																									
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																									
7 5/ T_C = 25°C	Truth table tests	3014	55	B	B	B	B	B	H	H	GND	H	H	B	B	B	B	B	B	B	B	5.0 V	All outputs	6/	6/	
			56	B	A	-	-	A	H	L	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			57	A	-	-	-	A	L	H	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			58	-	B	-	-	A	H	H	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			59	B	-	-	-	A	L	L	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			60	-	B	-	-	A	H	L	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			61	B	-	-	-	A	L	H	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			62	-	A	-	-	A	H	H	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			63	A	-	-	-	A	L	L	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			64	-	A	-	-	A	H	L	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			65	A	-	-	-	A	L	H	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			66	-	-	B	-	B	H	H	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			67	B	-	B	-	B	L	L	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			68	-	-	B	-	B	H	L	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			69	B	-	B	-	B	L	H	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			70	-	A	B	-	B	H	H	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			71	A	-	B	-	B	L	L	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			72	-	A	B	-	B	H	L	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			73	A	-	B	-	B	L	H	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			74	-	A	B	-	B	H	H	-	-	L	B	-	-	-	A	-	-	A	-	-	-	-	
			75	A	-	B	-	B	L	L	-	-	L	A	-	-	-	B	-	-	B	-	-	-	-	
			76	-	-	-	B	A	H	L	-	-	H	B	-	-	-	A	-	-	A	-	-	-	-	
			77	B	-	-	B	A	L	H	-	-	H	A	-	-	-	B	-	-	B	-	-	-	-	
			78	-	B	-	B	A	H	H	-	-	H	B	-	-	-	A	-	-	A	-	-	-	-	
			79	B	A	-	B	A	L	L	-	-	H	A	-	-	-	B	-	-	B	-	-	-	-	
			80	-	-	B	B	A	H	L	-	-	H	B	-	-	-	A	-	-	A	-	-	-	-	
			81	B	-	B	B	A	L	H	-	-	H	A	-	-	-	B	-	-	B	-	-	-	-	
8 5/	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and -55°C.																									

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open)

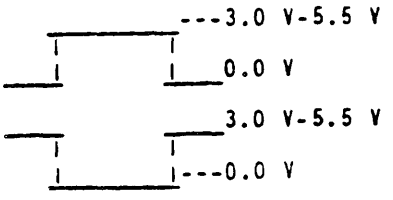
Subgroup	Symbol	MIL-STD-883 method	Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20	Measured terminal	Limits		Unit
			Cases A, B, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1CLR	1D	1CLK	1PR	1Q	1$\overline{Q}$	GND	2$\overline{Q}$	2Q	2PR	2CLK	2D	2CLR	V_{CC}				
9 T_C = 25°C	f_{MAX} 7/	Fig. 5	82	3.5 V	IN	IN	3.5 V	OUT		GND							5.0 V	1CLK to 1Q	30		MHz
		"	83	3.5 V	IN	IN	3.5 V		OUT	"							"	1CLK to 1$\overline{Q}$	"		"
		"	84							"	OUT		3.5 V	IN	IN	3.5 V	"	2CLK to 2Q	"		"
		"	85							"		OUT	3.5 V	IN	IN	3.5 V	"	2CLK to 2$\overline{Q}$	"		"
	t_{PLH1}	3003	86							"			IN			IN	"	2PR to 2Q	3	13	ns
		"	87							"	OUT	OUT	IN			IN	"	2CLR to 2$\overline{Q}$	"	"	"
		"	88	IN			IN		OUT	"							"	1CLR to 1$\overline{Q}$	"	"	"
		"	89	IN			IN	OUT		"							"	1PR to 1Q	"	"	"
	t_{PHL1}	"	90	IN			IN	OUT		"							"	1CLR to 1Q	5	15	"
		"	91	IN			IN		OUT	"							"	1PR to 1$\overline{Q}$	"	"	"
		"	92							"	OUT		IN			IN	"	2PR to 2$\overline{Q}$	"	"	"
		"	93							"		OUT	IN			IN	"	2CLR to 2Q	"	"	"
	t_{PLH2}	"	94	3.5 V	IN	IN	3.5 V	OUT		"							"	1CLK to 1Q	"	16	"
		"	95	3.5 V	IN	IN	3.5 V		OUT	"							"	1CLK to 1$\overline{Q}$	"	"	"
		"	96							"	OUT		3.5 V	IN	IN	3.5 V	"	2CLK to 2Q	"	"	"
		"	97							"		OUT	3.5 V	IN	IN	3.5 V	"	2CLK to 2$\overline{Q}$	"	"	"
	t_{PHL2}	"	98	3.5 V	IN	IN	3.5 V	OUT		"							"	1CLK to 1Q	"	18	"
		"	99	3.5 V	IN	IN	3.5 V		OUT	"							"	1CLK to 1$\overline{Q}$	"	"	"
		"	100							"	OUT		3.5 V	IN	IN	3.5 V	"	2CLK to 2Q	"	"	"
		"	101							"		OUT	3.5 V	IN	IN	3.5 V	"	2CLK to 2$\overline{Q}$	"	"	"
10	f_{MAX} 7/	"	102-105	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.															30		MHz
	t_{PLH1}	"	106-109																3	15	ns
	t_{PHL1}	"	110-113																5	17	"
	t_{PLH2}	"	114-117																5	18	"
	t_{PHL2}	"	118-121																7	20	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

See footnotes at end of device type 01.

MIL-M-38510/327A

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1/ Pins not referenced are N/C.



2/

3/

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

5/ Tests shall be performed in sequence, attributes data only.

6/ Output voltages shall be either: (1) H $\geq$2.5 V minimum and L $\leq$0.4 V maximum when using a high-speed checker double comparator; (2) H $\geq$1.5 V and L $\leq$1.5 V when using a high-speed checker single comparator.

7/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

8/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (µA) for circuit		
	A	B	C
I_{IL1}	0/-200	0/-200	0/-200
I_{IL4}	0/-400	0/-400	0/-400

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	1 1CLR	2 1J	3 1K	4 1CLK	5 1PR	6 1Q	7 1Q	8 GND	9 2Q	10 2Q	11 1PR	12 2CLK	13 2K	14 2J	15 2CLR	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	2.0 V	0.8 V	0.8 V	GND	0.8 V	-0.4 mA		GND								4.5 V	1Q	2.4		V
			2	0.8 V	0.8 V	0.8 V	GND	2.0 V		-0.4 mA	"								"	1Q	"		"
			3	2.0 V	2.0 V	2.0 V	2/	"	-0.4 mA		"								"	1Q	"		"
			4	2.0 V	0.8 V	0.8 V	2/	"		-0.4 mA	"								"	1Q	"		"
			5								"	-0.4 mA		2.0 V	GND	2.0 V	2.0 V	0.8 V	"	2Q	"		"
			6								"		-0.4 mA	0.8 V	GND	2.0 V	2.0 V	0.8 V	"	2Q	"		"
			7								"	-0.4 mA		2.0 V	2/	0.8 V	0.8 V	"	"	2Q	"		"
			8								"		-0.4 mA	2.0 V	2/	2.0 V	2.0 V	"	"	2Q	"		"
	V_{OL}	3007	9	0.8 V	0.8 V	0.8 V	GND	2.0 V	4 mA		"								"	1Q		0.4	"
			10	2.0 V	"	"	GND	0.8 V		4 mA	"								"	1Q		"	"
			11	"	"	"	2/	2.0 V	4 mA		"								"	1Q		"	"
			12	"	2.0 V	2.0 V	"	2.0 V		4 mA	"								"	1Q		"	"
			13								"	4 mA		0.8 V	GND	0.8 V	0.8 V	2.0 V	"	2Q		"	"
			14								"		4 mA	2.0 V	GND	0.8 V	0.8 V	0.8 V	"	2Q		"	"
			15								"	4 mA		"	2/	2.0 V	2.0 V	2.0 V	"	2Q		"	"
			16								"		4 mA	"	2/	0.8 V	0.8 V	2.0 V	"	2Q		"	"
	V_{IC}		17	-18 mA							"								"	1CLR		-1.5	"
			18		-18 mA						"								"	1J		"	"
			19			18 mA					"								"	1K		"	"
			20				18 mA				"								"	1CLK		"	"
			21					-18 mA			"								"	1PR		"	"
			22								"				-18 mA				"	2PR		"	"
			23								"				-18 mA				"	2CLK		"	"
			24								"					-18 mA			"	2K		"	"
			25								"						-18 mA		"	2J		"	"
			26								"							-18 mA	"	2CLR		"	"
	I_{IL2}	3009	27	3/	0.4 V	5.0 V	GND	5.0 V			"			3/	GND	0.4 V	5.0 V	5.0 V	5.5 V	1J	-8/	-8/	mA
			28	5.0 V	5.0 V	0.4 V	GND	3/			"			5.0 V	GND	5.0 V	0.4 V	3/	"	1K	"	"	"
			29								"								"	2K	"	"	"
			30								"								"	2J	"	"	"
			31	3/	5.0 V	5.0 V	0.4 V	5.0 V			"			5.0 V	0.4 V	5.0 V	5.0 V	3/	"	1CLK	"	"	"
			32								"			5.0 V	0.4 V	5.0 V	5.0 V	3/	"	2CLK	"	"	"
	I_{IL4}	"	33	0.4 V	5.0 V	5.0 V	5.0 V	GND			"								"	1CLR	"	"	"
		"	34	GND	5.0 V	5.0 V	5.0 V	0.4 V			"								"	1PR	"	"	"
		"	35								"			0.4 V	5.0 V	5.0 V	5.0 V	GND	"	2PR	"	"	"
		"	36								"			GND	5.0 V	5.0 V	5.0 V	0.4 V	"	2CLR	"	"	"
	I_{IH2}	3010	37	GND	2.7 V	GND	GND	GND			"								"	1J		20	μA
		"	38	GND	GND	2.7 V	GND	GND			"								"	1K		"	"
		"	39								"			GND	GND	2.7 V	GND	GND	"	2K		"	"
		"	40								"			GND	GND	GND	2.7 V	GND	"	2J		"	"
	I_{IH4}	"	41	2.7 V	GND	GND	GND	GND			"								"	1CLR		10	"
		"	42	GND	4.5 V	GND	GND	2.7 V			"								"	1PR		"	"
		"	43								"			2.7 V	GND	GND	4.5 V	GND	"	2PR		"	"
		"	44								"			GND	GND	GND	GND	2.7 V	"	2CLR		"	"
	I_{IH5}	"	45	GND	GND	GND	2.7 V	GND			"								"	1CLK		20	"
		"	46								"			GND	2.7 V	GND	GND	GND	"	2CLK		"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	2 1 1CLR	3 2 1J	4 3 1K	5 4 1CLK	7 5 1PR	8 6 1Q	9 7 1Q	10 8 GND	12 9 2Q	13 10 2Q	14 11 2PR	15 12 2CLK	17 13 2K	18 14 2J	19 15 2CLR	20 16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{IH7}	3010	47	GND	7.0 V	GND	GND	GND			GND			GND	GND	7.0 V	GND	GND	5.5 V	1J		100	µA
			48	GND	GND	7.0 V	GND	GND			-			GND	GND	GND	7.0 V	GND	-	1K		-	-
			49	-	-	-	-	-			-			-	-	-	-	-	-	2K		-	-
			50	-	-	-	-	-			-			-	-	-	-	-	-	2J		-	-
	I_{IH0}	-	51	7.0 V	GND	GND	GND	GND			-			7.0 V	GND	GND	GND	GND	-	1CLR		200	-
		-	52	GND	GND	GND	GND	7.0 V			-			GND	GND	GND	GND	7.0 V	-	1PR		-	-
		-	53	-	-	-	-	-			-			-	-	-	-	-	-	2PR		-	-
		-	54	-	-	-	-	-			-			-	-	-	-	-	-	2CLR		-	-
	I_{IH10}	-	55	GND	GND	GND	7.0 V	GND			-			GND	7.0 V	GND	GND	GND	-	1CLK		100	-
		-	56	-	-	-	-	-			-			-	-	-	-	-	-	2CLK		-	-
	I_O 4/	3011	57	5.0 V				GND	2.25 V	2.25 V	-	2.25 V	2.25 V	5.0 V				GND	-	1Q	-30	-112	-
		-	58	GND				5.0 V			-			GND				5.0 V	-	1Q	-	-	-
		-	59	-				-			-			-				-	-	2Q	-	-	-
		-	60	-				-			-			-				-	-	2Q	-	-	-
	I_{CC}	3005	61	5.5 V	GND	GND	GND	GND			-			GND	GND	GND	GND	5.5 V	-	V_{CC}		4	mA
		-	62	GND	GND	GND	GND	5.5 V			-			5.5 V	GND	GND	GND	GND	-	V_{CC}		-	-
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C.																						
7 5/ T_C = 25°C	Truth table tests	3014	63	A	A	A	B	B	H	L	GND	L	H	B	B	A	A	A	5.0 V	All outputs	6/	6/	
		-	64	-	-	-	A	A	-	-	-	-	-	-	A	-	-	-	-	-	-	-	
		-	65	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	66	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	67	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	68	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	69	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	70	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	71	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	72	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	73	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	75	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	76	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	77	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	78	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	79	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	80	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	81	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	82	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	83	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	84	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	85	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	86	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	87	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	88	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	89	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	90	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	91	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	92	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	93	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	94	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	95	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	96	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	97	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	98	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
		-	99	D	A	B	-	A	A	H	H	H	L	B	A	B	B	B	-	-	-	-	
		-	100	A	A	-	A	B	A	H	H	L	H	A	A	A	A	A	-	-	-	-	
		-	101	A	B	-	B	A	A	L	L	H	L	A	A	A	A	B	-	-	-	-	
		-	102	B	-	-	-	-	A	L	L	H	L	A	A	A	A	A	-	-	-	-	

See footnotes at end of device type 02.

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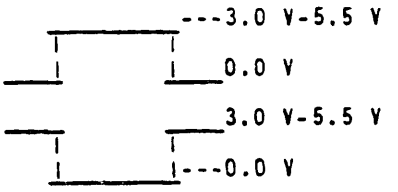
TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	1 1$\overline{CLR}$	2 1J	3 1$\overline{K}$	4 1CLK	5 1$\overline{PR}$	6 1Q	7 1$\overline{Q}$	8 GND	9 2$\overline{Q}$	10 2Q	12 2$\overline{PR}$	13 2CLK	14 2$\overline{K}$	15 2J	16 2$\overline{CLR}$	17 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
8 5/	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = 25°C	f_{MAX} 7/	Fig. 5	103	3.5 V	3.5 V	GND	IN	3.5 V	OUT	OUT	GND	OUT	OUT	3.5 V	IN	GND	3.5 V	3.5 V	5.0 V	1Q	30		MHz
		"	104	3.5 V	3.5 V	GND	IN	3.5 V	OUT	OUT	"	OUT	OUT	3.5 V	IN	GND	3.5 V	3.5 V	"	1$\overline{Q}$	30		"
		"	105								"								"	2Q	30		"
		"	106								"								"	2$\overline{Q}$	30		"
	t_{PLH1}	3003 Fig. 5	107	IN	3.5 V	GND	IN	3.5 V	OUT	OUT	"	OUT	OUT	IN	IN	GND	GND	3.5 V	"	1$\overline{CLR}$ to 1Q	3.0	13.0	ns
		"	108	3.5 V	GND	GND	IN	3.5 V	OUT	OUT	"	OUT	OUT	3.5 V	IN	3.5 V	3.5 V	IN	"	1$\overline{PR}$ to 1Q	"	"	"
		"	109								"								"	2$\overline{PR}$ to 2Q	"	"	"
		"	110								"								"	2$\overline{CLR}$ to 2Q	"	"	"
	t_{PHL1}	"	111	IN	3.5 V	GND	IN	3.5 V	OUT	OUT	"	OUT	OUT	IN	IN	GND	GND	3.5 V	"	1$\overline{CLR}$ to 1$\overline{Q}$	5.0	15.0	"
		"	112	3.5 V	GND	GND	IN	IN	OUT	OUT	"	OUT	OUT	3.5 V	IN	3.5 V	3.5 V	IN	"	1$\overline{PR}$ to 1$\overline{Q}$	"	"	"
		"	113								"								"	2$\overline{PR}$ to 2$\overline{Q}$	"	"	"
		"	114								"								"	2$\overline{CLR}$ to 2$\overline{Q}$	"	"	"
	t_{PLH2}	"	115	9/	3.5 V	GND	IN	3.5 V	OUT	OUT	"	OUT	OUT	9/	IN	GND	3.5 V	3.5 V	"	1$\overline{CLR}$ to 1Q	"	16.0	"
		"	116	3.5 V	3.5 V	GND	IN	9/	OUT	OUT	"	OUT	OUT	3.5 V	IN	GND	3.5 V	9/	"	1$\overline{CLK}$ to 1$\overline{Q}$	"	"	"
		"	117								"								"	2$\overline{CLR}$ to 2Q	"	"	"
		"	118								"								"	2$\overline{CLK}$ to 2$\overline{Q}$	"	"	"
	t_{PHL2}	"	119	3.5 V	3.5 V	GND	IN	9/	OUT	OUT	"	OUT	OUT	3.5 V	IN	GND	3.5 V	9/	"	1$\overline{CLK}$ to 1Q	"	18.0	"
		"	120	9/	"	"	"	3.5 V	"	"	"	"	"	9/	IN	GND	3.5 V	3.5 V	"	1$\overline{CLK}$ to 1$\overline{Q}$	"	"	"
		"	121								"								"	2$\overline{CLK}$ to 2Q	"	"	"
		"	122								"								"	2$\overline{CLK}$ to 2$\overline{Q}$	"	"	"
10	f_{MAX}	3003 Fig. 5	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																		30		MHz
	t_{PLH2}	"																			3	15	ns
	t_{PHL1}	"																			5	17	"
	t_{PLH2}	"																			5	18	"
	t_{PHL2}	"																			7	20	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 02.

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1/ Pins not referenced are N/C.



2/

3/

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

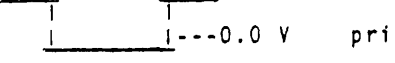
5/ Tests shall be performed in sequence, attributes data only.

6/ Output voltages shall be either: (1) H $\geq$2.5 V minimum and L $\leq$0.4 V maximum when using a high-speed checker double comparator; (2) H $\geq$1.5 V and L $\leq$1.5 V when using a high-speed checker single comparator.

7/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

8/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (uA) for circuit		
	A	B	C
I_{IL2}	0/-200	0/-200	0/-200
I_{IL4}	0/-400	0/-400	0/-400

9/ 

prior to pulse test.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 / Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1CLK	1K	1J	1$\overline{PR}$	1Q	1$\overline{Q}$	2$\overline{Q}$	GND	2Q	2$\overline{PR}$	2J	2K	2CLK	2$\overline{CLR}$	1$\overline{CLR}$	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1	2.0 V	2.0 V	2.0 V	0.8 V	-0.4 mA			GND							2.0 V	4.5 V	1Q	2.5		V
			2	2.0 V	2.0 V	"	2.0 V	"	-0.4 mA		"							0.8 V	"	1$\overline{Q}$			
			3	2/	0.8 V		"	-0.4 mA										2.0 V		1Q			
			4	2/	2.0 V	0.8 V	"	"	-0.4 mA										2.0 V	1$\overline{Q}$			
			5							-0.4 mA			2.0 V	2.0 V	2.0 V	2.0 V	0.8 V		"	2Q			
			6							-0.4 mA		-0.4 mA	0.8 V	2.0 V	"	2.0 V	2.0 V		"	2$\overline{Q}$			
			7										2.0 V	0.8 V	0.8 V	2/	"		"	2Q			
			8									-0.4 mA	2.0 V	2.0 V		2/				2$\overline{Q}$			
	V_{OL}	3007	9	2.0 V	2.0 V	2.0 V	0.8 V	4 mA	4 mA										2.0 V	1Q		0.4	
			10	2.0 V	2.0 V	"	2.0 V												0.8 V	1$\overline{Q}$			
			11	2/	0.8 V		"	4 mA											2.0 V	1Q			
			12	2/	2.0 V	0.8 V	"												"	1$\overline{Q}$			
			13							4 mA			0.8 V	2.0 V	2.0 V	2.0 V	2.0 V			2Q			
			14									4 mA	2.0 V	2.0 V	"	2.0 V	0.8 V			2$\overline{Q}$			
			15										"	0.8 V	"	2/	2.0 V			2Q			
			16							4 mA			"	2.0 V	0.8 V	2/	2.0 V			2$\overline{Q}$			
	V_{IC}		17	-18 mA																1CLK		-1.5	
			18		-18 mA															1K			
			19			-18 mA														1J			
			20				-18 mA													1PR			
			21										-18 mA							2$\overline{PR}$			
			22											-18 mA						2J			
			23												-18 mA					2K			
			24													-18 mA				2CLK			
			25														-18 mA			2$\overline{CLR}$			
			26															-18 mA		1$\overline{CLR}$			
	I_{IL2}	3009	27	5.0 V	0.4 V	5.0 V	3/											5.0 V	5.5 V	1K	8/	8/	uA
			28	5.0 V	5.0 V	0.4 V	5.0 V											3/	"	1J	"	"	
			29										5.0 V	0.4 V	5.0 V	5.0 V	3/		"	2J	"	"	
			30										3/	5.0 V	0.4 V	5.0 V	5.0 V		"	2K	"	"	
			31	0.4 V	5.0 V	5.0 V	3/											5.0 V	"	1CLK	"	"	
			32	0.4 V	5.0 V	5.0 V	5.0 V											3/	"	1CLK	"	"	
	I_{IL4}	"	33	5.0 V	5.0 V	5.0 V	0.4 V											5.0 V	"	1PR	"	"	
		"	34	5.0 V	5.0 V	5.0 V	5.0 V											0.4 V	"	1$\overline{CLR}$	"	"	
		"	35										0.4 V	5.0 V	5.0 V	5.0 V	5.0 V		"	2$\overline{PR}$	"	"	
	I_{IL2}	"	36										3/	5.0 V	5.0 V	0.4 V	5.0 V		"	2CLK	"	"	
		"	37										5.0 V	5.0 V	5.0 V	0.4 V	3/		"	2CLK	"	"	
	I_{IL4}	"	38										5.0 V	5.0 V	5.0 V	5.0 V	0.4 V		"	2$\overline{CLR}$	"	"	
	I_{IH2}	3010	39	GND	2.7 V	5.0 V	GND											5.0 V	"	1K		20	
		"	40	GND	5.0 V	2.7 V	5.0 V											GND	"	1J		"	
		"	41										5.0 V	2.7 V	5.0 V	GND	GND		"	2J		"	
		"	42										GND	5.0 V	2.7 V	GND	5.0 V		"	2K		"	
	I_{IH4}	"	43	GND	GND	5.0 V	2.7 V											3/	"	1PR		40	
		"	44										2.7 V	5.0 V	GND	GND	3/		"	2$\overline{PR}$		"	
		"	45										3/	GND	5.0 V	GND	2.7 V		"	2$\overline{CLR}$		"	
		"	46	GND	5.0 V	GND	3/											2.7 V	"	1$\overline{CLR}$		"	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	1 1CLK	2 1K	3 1J	4 1PR	5 1Q	6 1Q	7 GND	8 2Q	9 2Q	10 2J	11 2K	12 2PR	13 2CLK	14 2CLK	15 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{IH5}	3010	47 48	2.7 V	GND	GND	GND					GND		GND	GND	GND	2.7 V	GND	GND	5.5 V	1CLK 2CLK		10	µA
	I_{IH7}	"	49 50 51 52	GND GND	7.0 V 5.0 V	5.0 V 7.0 V	GND 5.0 V							5.0 V GND	7.0 V 5.0 V	5.0 V 7.0 V	GND GND	GND 5.0 V	5.0 V GND	"	1K 1J 2J 2K		100	"
	I_{IH9}	"	53 54 55 56	GND GND	GND 5.0 V	5.0 V GND	7.0 V 3/							7.0 V 3/	5.0 V 7.0 V	GND 5.0 V	GND GND	3/ 7.0 V	3/ 7.0 V	"	1PR 2PR 2CLK 1CLK		200	"
	I_{IH10}	"	57 58	7.0 V	GND	GND	GND							GND	GND	GND	7.0 V	GND	GND	"	1CLK 2CLK		100	"
	I_O 4/	3011	59 60 61 62				GND 5.0 V	2.5 V	2.25 V	2.25 V		2.25 V		5.0 V GND				GND 5.0 V	5.0 V GND	"	1Q 1Q 2Q 2Q	9/	9/	mA
	I_{CC}	3005	63 64	GND GND	GND GND	GND GND	GND 5.5 V							GND 5.5 V	GND GND	GND GND	GND GND	5.5 V GND	5.5 V GND	"	V_{CC} V_{CC}		45	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IL} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IL} tests are omitted.																					
7 T_C = 25°C	Truth table tests	3014	65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93	B A B A B B A B A B A B A B A B A B A B A B A B A B A B A B	B A A B A B B A B A B A B A B A B A B A B A B A B A B A B A	A B B A B A A B A B A B A B A B A B A B A B A B A B A B A B	A B B A B A A B A B A B A B A B A B A B A B A B A B A B A B	L H H L H L L H L H L H L H L H L H L H L H L H L H L H L	H L L H L H H L H L H L H L H L H L H L H L H L H L H L H	H L L H L H H L H L H L H L H L H L H L H L H L H L H L H	GND	L H H L H L L H L H L H L H L H L H L H L H L H L H L H L	A B B A B A A B A B A B A B A B A B A B A B A B A B A B A B	A B B A B A A B A B A B A B A B A B A B A B A B A B A B A B	B A A B A B B A B A B A B A B A B A B A B A B A B A B A B A	B A B A B A A B A B A B A B A B A B A B A B A B A B A B A B	B A A B A B B A B A B A B A B A B A B A B A B A B A B A B A	B A B A B A A B A B A B A B A B A B A B A B A B A B A B A B	5.0 V	All outputs	6/	6/	

See footnotes at end of device type 03.

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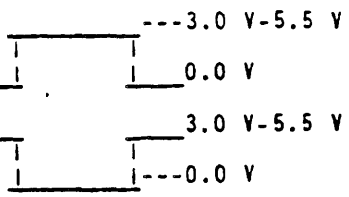
TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 / Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	1CLK	1K	1J	$1\overline{PR}$	1Q	$1\overline{Q}$	$2\overline{Q}$	GND	2Q	$2\overline{PR}$	2J	2K	2CLK	$2\overline{CLR}$	$1\overline{CLR}$	V_{CC}				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = 25°C	f_{MAX} 7/	Fig. 5	94	IN	3.5 V	3.5 V	3.5 V	OUT	OUT		GND								5.0 V	1Q	25		MHz
		"	95	IN	3.5 V	3.5 V	3.5 V				"							3.5 V	"	$1\overline{Q}$				
		"	96							OUT	"								"	$2\overline{Q}$				
		"	97								"	OUT	3.5 V	3.5 V	3.5 V	IN	3.5 V		"	2Q				
	t_{PLH1}	3003	98	IN	3.5 V	GND	IN	OUT	OUT		"							3.5 V	"	1$\overline{PR}$ to 1Q	3	15	ns
		"	99	IN	GND	3.5 V	3.5 V				"							IN	"	1$\overline{CLR}$ to $1\overline{Q}$	"	"		
		"	100							OUT	"								"	2$\overline{CLR}$ to 2Q	"	"		
		"	101								"	OUT	3.5 V	3.5 V	GND	IN	IN		"	2$\overline{PR}$ to 2Q	"	"		
	t_{PHL1}	"	102	IN	3.5 V	GND	IN	OUT	OUT		"							3.5 V	"	1$\overline{PR}$ to $1\overline{Q}$	4	18	"
		"	103	IN	GND	3.5 V	3.5 V				"							IN	"	1$\overline{CLR}$ to 1Q	"	"		
		"	104							OUT	"								"	2$\overline{CLR}$ to 2Q	"	"		
		"	105								"	OUT	3.5 V	3.5 V	GND	IN	IN		"	2$\overline{PR}$ to 2Q	"	"		
	t_{PLH2}	"	106	IN	3.5 V	3.5 V	3.5 V	OUT	OUT		"							3.5 V	"	1CLK to $1\overline{Q}$	3	15	"
		"	107	IN	3.5 V	3.5 V	3.5 V				"							3.5 V	"	1CLK to 1Q	"	"		
		"	108							OUT	"								"	2CLK to $2\overline{Q}$	"	"		
		"	109								"	OUT	3.5 V	3.5 V	3.5 V	IN	3.5 V		"	2CLK to 2Q	"	"		
	t_{PHL2}	"	110	IN	3.5 V	3.5 V	3.5 V	OUT	OUT		"							3.5 V	"	1CLK to $1\overline{Q}$	5	19	"
		"	111	IN	3.5 V	3.5 V	3.5 V				"							3.5 V	"	1CLK to 1Q	"	"		
		"	112							OUT	"								"	2CLK to $2\overline{Q}$	"	"		
		"	113								"	OUT	3.5 V	3.5 V	3.5 V	IN	3.5 V		"	2CLK to 2Q	"	"		
10 T_C = 125°C	f_{MAX} 7/	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																			25		MHz
	t_{PLH1}																				3	20	ns
	t_{PHL1}																				4	22	"
	t_{PLH2}																				3	18	"
	t_{PHL2}																				5	23	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 03.

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1/ Pins not referenced are N/C.



2/

3/

4/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

5/ Tests shall be performed in sequence, attributes data only.

6/ Output voltages shall be either: (1) H $\geq$2.5 V minimum and L $\leq$0.4 V maximum when using a high-speed checker double comparator; (2) H $\geq$1.5 V and L $\leq$1.5 V when using a high-speed checker single comparator.

7/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

8/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (µA) for circuit		
	A	B	C
I_{IL2}	0/-200	0/-200	0/-200
I_{IL4}	0/-400	0/-400	0/-400

9/ I_O limits are as follows:

Test	Min/Max limits in (mA) for circuit		
	A	B	C
I_O	-30/-112	-30/-112	-30/-112

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, 8, 5	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
T_C = 25°C	V_{OH}	3006	1	IOC	10.8 V	12.0 V	12.0 V	12.0 V	12.0 V	12.0 V	12.0 V	12.0 V	GND	1/	1.0 mA	1.0 mA	1.0 mA	1.0 mA	1.0 mA	1.0 mA	1.0 mA	1.0 mA	1.0 mA	1.0 mA	10	2.4		V
			2																						20			
			3																						30			
			4																						40			
			5																						50			
			6																						60			
			7																						70			
			8																						80			
	V_{OL}	3007	9		10.9 V	10.8 V	10.8 V	10.8 V	10.8 V	10.8 V	10.8 V	10.8 V				1.0 mA		12 mA		12 mA	12 mA	12 mA	12 mA	12 mA	10		0.4	V
			10																						20			
			11																						30			
			12																						40			
			13																						50			
			14																						60			
			15																						70			
			16																						80			
	V_{IC}		17		-1.18 mA	-1.18 mA	-1.18 mA	-1.18 mA	-1.18 mA	-1.18 mA	-1.18 mA	-1.18 mA			-1.19 mA										CLK	-1.5		V
			18																						10			
			19																						20			
			20																						30			
			21																						40			
			22																						50			
			23																						60			
			24																						70			
			25																						80			
			26																						CLK			
	I_{IL1}	1009	27		10.4 V	10.4 V	10.4 V	10.4 V	10.4 V	10.4 V	10.4 V	10.4 V			10.4 V										DC	5.5 V		μA
			28																						10			
			29																						20			
			30																						30			
			31																						40			
			32																						50			
			33																						60			
			34																						70			
			35																						80			
			36																						CLK			
	I_{IL2}	1010	37		12.7 V	12.7 V	12.7 V	12.7 V	12.7 V	12.7 V	12.7 V	12.7 V			12.7 V										DC	R		μA
			38																						10			
			39																						20			
			40																						30			
			41																						40			
			42																						50			
			43																						60			
			44																						70			
			45																						80			
			46																						CLK			
	I_{IL3}		47		17.0 V	17.0 V	17.0 V	17.0 V	17.0 V	17.0 V	17.0 V	17.0 V			17.0 V										DC	100		μA
			48																						10			
			49																						20			
			50																						30			
			51																						40			
			52																						50			
			53																						60			
			54																						70			
			55																						80			
			56																						CLK			

1/ See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases Q, R, S Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				OC	1O	2O	3O	4O	5O	6O	7O	8O	GND	CLK	9O	7O	6O	5O	4O	3O	2O	1O	V_{CC}				
1 T_C = 25°C	I_O 2/	3011	57	GND	5.0 V								GND	1/							12.25 V	12.25 V	5.5 V	1O	-15	-110	mA
			58			5.0 V												12.5 V						2O			
			59				5.0 V												12.25 V					3O			
			60					5.0 V												12.25 V				4O			
			61						5.0 V															5O			
			62							5.0 V						12.25 V								6O			
			63								5.0 V				12.25 V									7O			
			64									5.0 V												8O			
	I_{OZH}		55	5.0 V	12.0 V																			1O		20	µA
			66			12.0 V												12.7 V				12.7 V		2O			
			67				12.0 V												12.7 V					3O			
			68					12.0 V												12.7 V				4O			
			69						12.0 V												12.7 V			5O			
			70							12.0 V						12.7 V								6O			
			71								12.0 V				12.7 V									7O			
			72									12.0 V												8O			
	I_{OZL}		73	5.0 V	10.8 V																			1O		-20	
			74			10.8 V												10.4 V				10.4 V		2O			
			75				10.8 V												10.4 V					3O			
			76					10.8 V												10.4 V				4O			
			77						10.8 V												10.4 V			5O			
			78							10.8 V						10.4 V								6O			
			79								10.8 V				10.4 V									7O			
			80									10.8 V												8O			
	I_{CCH}	3005	81	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	-	1/									-	V_{CC}		17	mA
	I_{CCL}	"	82	"	GND	GND	GND	GND	GND	GND	GND	GND	-	-									-	V_{CC}		23	"
	I_{CCZ}	"	83	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	-	-									-	V_{CC}		27	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 3/ T_C = 25°C	Truth table tests	3014	84	B	A	A	A	A	A	A	A	A	GND	L	X	X	X	X	X	X	X	X	5.0 V	All outputs	4/	8/	
			85		A	B	A	A	A	A	A	A		H	H	H	H	H	H	H	H	H					
			86		B	A	B	A	A	A	A	A		L	H	L	H	H	H	H	H	H					
			87		B	B	A	B	A	A	A	A		H	L	H	L	H	H	H	H	H					
			88		A	A	B	B	A	A	A	A		L	L	L	L	H	H	H	H	H					
			89		A	A	A	A	B	B	A	A		H	H	H	H	L	L	L	L	H					
			90		A	A	A	A	A	A	B	B		L	H	L	H	L	L	L	L	H					
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = 25°C	f_{MAX} 5/	Fig. 5	91	GND	IN									IN										1O	10		MHz
			92			IN																		2O			
			93				IN																	3O			
			94					IN																4O			
			95						IN															5O			
			96							IN						OUT								6O			
			97								IN						OUT							7O			
			98									IN						OUT						8O			

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases I2, R, S1 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured	Limits		Unit
				$\overline{OE}$	1D	2D	3D	4D	5D	6D	7D	8D	GND	CLK	8Q	7Q	6Q	5Q	4Q	3Q	2Q	1Q	V_{CC}	terminal	Min	Max	
9 T_C = 25°C	t_{PLH2}	3003 Fig. 5	99	GND	IN								GND	IN									5.0 V	tCLK to 1Q	4	12	ns
			100	"		IN							"	"								OUT	"	tCLK to 2Q	"	"	"
			101	"			IN						"	"						OUT			"	tCLK to 3Q	"	"	"
			102	"				IN					"	"					OUT				"	tCLK to 4Q	"	"	"
			103	"					IN				"	"				OUT					"	tCLK to 5Q	"	"	"
			104	"						IN			"	"			OUT						"	tCLK to 6Q	"	"	"
			105	"							IN		"	"		OUT							"	tCLK to 7Q	"	"	"
			106	"								IN	"	"	OUT								"	tCLK to 8Q	"	"	"
	t_{PHL2}	"	107	"	IN								"	"								OUT	"	tCLK to 1Q	"	"	"
			108	"		IN							"	"							OUT		"	tCLK to 2Q	"	"	"
			109	"			IN						"	"						OUT			"	tCLK to 3Q	"	"	"
			110	"				IN					"	"					OUT				"	tCLK to 4Q	"	"	"
			111	"					IN				"	"				OUT					"	tCLK to 5Q	"	"	"
			112	"						IN			"	"			OUT						"	tCLK to 6Q	"	"	"
			113	"							IN		"	"		OUT							"	tCLK to 7Q	"	"	"
			114	"								IN	"	"	OUT								"	tCLK to 8Q	"	"	"
	t_{PZL}	"	115	IN	GND								"	1/								OUT	"	$\overline{OE}$ to 1Q	"	18	"
			116	"		GND							"	"							OUT		"	$\overline{OE}$ to 2Q	"	"	"
			117	"			GND						"	"						OUT			"	$\overline{OE}$ to 3Q	"	"	"
			118	"				GND					"	"					OUT				"	$\overline{OE}$ to 4Q	"	"	"
			119	"					GND				"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"
			120	"						GND			"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"
			121	"							GND		"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"
			122	"								GND	"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"
	t_{PZH}	"	123	"	5.0 V								"	"							OUT		"	$\overline{OE}$ to 1Q	"	"	"
			124	"		5.0 V							"	"								OUT	"	$\overline{OE}$ to 2Q	"	"	"
			125	"			5.0 V						"	"							OUT		"	$\overline{OE}$ to 3Q	"	"	"
			126	"				5.0 V					"	"						OUT			"	$\overline{OE}$ to 4Q	"	"	"
			127	"					5.0 V				"	"					OUT				"	$\overline{OE}$ to 5Q	"	"	"
			128	"						5.0 V			"	"				OUT					"	$\overline{OE}$ to 6Q	"	"	"
			129	"							5.0 V		"	"			OUT						"	$\overline{OE}$ to 7Q	"	"	"
			130	"								5.0 V	"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"
	t_{PLZ}	"	131	"	GND								"	"							OUT		"	$\overline{OE}$ to 1Q	3	13	"
			132	"		GND							"	"								OUT	"	$\overline{OE}$ to 2Q	"	"	"
			133	"			GND						"	"						OUT			"	$\overline{OE}$ to 3Q	"	"	"
			134	"				GND					"	"					OUT				"	$\overline{OE}$ to 4Q	"	"	"
			135	"					GND				"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"
			136	"						GND			"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"
			137	"							GND		"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"
			138	"								GND	"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"
	t_{PHZ}	"	139	"	5.0 V								"	"							OUT		"	$\overline{OE}$ to 1Q	2	8	"
			140	"		5.0 V							"	"								OUT	"	$\overline{OE}$ to 2Q	"	"	"
			141	"			5.0 V						"	"						OUT			"	$\overline{OE}$ to 3Q	"	"	"
			142	"				5.0 V					"	"					OUT				"	$\overline{OE}$ to 4Q	"	"	"
			143	"					5.0 V				"	"				OUT					"	$\overline{OE}$ to 5Q	"	"	"
			144	"						5.0 V			"	"			OUT						"	$\overline{OE}$ to 6Q	"	"	"
			145	"							5.0 V		"	"		OUT							"	$\overline{OE}$ to 7Q	"	"	"
			146	"								5.0 V	"	"	OUT								"	$\overline{OE}$ to 8Q	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S Test no.	1 $\overline{OE}$	2 1D	3 2D	4 3D	5 4D	6 5D	7 6D	8 7D	9 8D	10 GND	11 CLK	12 8Q	13 7Q	14 6Q	15 5Q	16 4Q	17 3Q	18 2Q	19 1Q	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = 25°C	f_{MAX}																								30		MHz
	t_{PLH2}																								6	15	ns
	t_{PHL7}																								*	*	*
	t_{P7L}		Same tests and terminal conditions as for subgroup 7, except T_C = +125°C.																						*	21	*
	t_{P7H}																								*	*	*
	t_{PL2}																								3	15	*
	t_{PH2}																								2	10	*
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ Apply ---3.0 V-5.5 V / -0.0 V pulse prior to test.

2/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

3/ Tests shall be performed in sequence, attributes data only.

4/ Output voltages shall be either: (1) H ≥2.4 V minimum and L ≤0.4 V maximum when using a high-speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high-speed checker single comparator.

5/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

6/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (µA) for circuit A	B	C
I_{IL}	0/-200	0/-200	0/-200

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S Test no.	1 $\overline{OC}$	2 1D	3 2D	4 3D	5 4D	6 5D	7 6D	8 7D	9 8D	10 GND	11 CLK	12 $8\overline{Q}$	13 $7\overline{Q}$	14 $6\overline{Q}$	15 $5\overline{Q}$	16 $4\overline{Q}$	17 $3\overline{Q}$	18 $2\overline{Q}$	19 $1\overline{Q}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
T_C = 25°C	V_{OH}	3006	1	0.8 V	0.8 V								GND	1/									4.5 V	1$\overline{Q}$	2.4		V
			2			0.8 V															-1.0 mA	-1.0 mA		2$\overline{Q}$			
			3				0.8 V													-1.0 mA				3$\overline{Q}$			
			4					0.8 V											-1.0 mA					4$\overline{Q}$			
			5						0.8 V									1.0 mA						5$\overline{Q}$			
			6							0.8 V							1.0 mA							6$\overline{Q}$			
			7								0.8 V					-1.0 mA								7$\overline{Q}$			
			8									0.8 V			-1.0 mA									8$\overline{Q}$			
	V_{OL}	3007	9		2.0 V																	12 mA		1$\overline{Q}$		0.4	
			10			2.0 V															12 mA			2$\overline{Q}$			
			11				2.0 V													12 mA				3$\overline{Q}$			
			12					2.0 V											12 mA					4$\overline{Q}$			
			13						2.0 V									12 mA						5$\overline{Q}$			
			14							2.0 V							12 mA							6$\overline{Q}$			
			15								2.0 V					12 mA								7$\overline{Q}$			
			16									2.0 V			12 mA									8$\overline{Q}$			
	V_{IC}		17	-18 mA																				$\overline{OC}$		-1.5	
			18		-18 mA																			1D			
			19			-18 mA																		2D			
			20				-18 mA																	3D			
			21					-18 mA																4D			
			22						-18 mA															5D			
			23							-18 mA														6D			
			24								-18 mA													7D			
			25									-18 mA												8D			
			26											-18 mA										CLK			
	I_{IL3}	3009	27	0.4 V																			5.5 V	$\overline{OC}$	6/	6/	μA
			28		0.4 V																			1D			
			29			0.4 V																		2D			
			30				0.4 V																	3D			
			31					0.4 V																4D			
			32						0.4 V															5D			
			33							0.4 V														6D			
			34								0.4 V													7D			
			35									0.4 V												8D			
			36											0.4 V										CLK			
	I_{IH3}	3010	37	2.7 V																				$\overline{OC}$		20	
			38		2.7 V																			1D			
			39			2.7 V																		2D			
			40				2.7 V																	3D			
			41					2.7 V																4D			
			42						2.7 V															5D			
			43							2.7 V														6D			
			44								2.7 V													7D			
			45									2.7 V												8D			
			46											2.7 V										CLK			
	I_{IH8}		47	7.0 V																				$\overline{OC}$		100	
			48		7.0 V																			1D			
			49			7.0 V																		2D			
			50				7.0 V																	3D			
			51					7.0 V																4D			
			52						7.0 V															5D			
			53							7.0 V														6D			
			54								7.0 V													7D			
			55									7.0 V												8D			
			56											7.0 V										CLK			

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases Q, R, S Test no.	1 $\overline{PE}$	2 $1\overline{D}$	3 $2\overline{D}$	4 $3\overline{D}$	5 $4\overline{D}$	6	7 $5\overline{D}$	8	9	10 GND	11 CLK	12 $6\overline{D}$	13 $7\overline{D}$	14 $8\overline{D}$	15 $9\overline{D}$	16 $10\overline{D}$	17 $11\overline{D}$	18 $12\overline{D}$	19 $13\overline{D}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{IO} 2/	3011	57	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	17	12.25 V	12.25 V	12.25 V	12.25 V	12.25 V	12.25 V	12.25 V	12.25 V	5.5 V	1Q	15	110	mA
			58																					2Q			
			59																					3Q			
			60																					4Q			
			61																					5Q			
			62																					6Q			
			63																					7Q			
			64																					8Q			
	I_{IOZH}		55	5.0 V	10.8 V																			1Q		20	µA
			66			10.8 V																		2Q			
			67				10.8 V																	3Q			
			68					10.8 V																4Q			
			69						10.8 V															5Q			
			70							10.8 V														6Q			
			71								10.8 V													7Q			
			72									10.8 V												8Q			
	I_{OZL}		73		12.0 V																			1Q		-20	
			74			12.0 V																		2Q			
			75				12.0 V																	3Q			
			76					12.0 V																4Q			
			77						12.0 V															5Q			
			78							12.0 V														6Q			
			79								12.0 V													7Q			
			80									12.0 V												8Q			
	I_{CCH}	3005	81	GND	GND	GND	GND	GND	GND	GND	GND	GND	-	-									-	V_{CC}		17	mA
	I_{CCL}	"	82	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	-	-									-	V_{CC}		73	"
	I_{CCZ}	"	83	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	-	-									-	V_{CC}		27	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 3/ T_C = 25°C	Truth table tests	3014	84	B	B	B	B	B	B	B	B	B	B	GND	A	X	X	X	X	X	X	X	X	5.0 V	All outputs	4/	4/	
			85		B	B	B	B	B	B	B	B	B		A	H	H	H	H	H	H	H	H				
			86		A	A	A	A	A	A	A	A	A		3	-	-	-	-	-	-	-	-				
			87		-	-	-	-	-	-	-	-	-		A	L	L	L	L	L	L	L	L				
			88		-	-	-	-	-	-	-	-	-		3	L	L	L	L	L	L	L	L				
			89		B	B	B	B	B	B	B	B	B		A	H	H	H	H	H	H	H	H				
			90		B	B	B	B	B	B	B	B	B		3	H	H	H	H	H	H	H	H				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = 25°C	f_{MAX} 5/	Fig. 5	91	GND	IN	IN	IN	IN	IN	IN	IN	IN	-	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V	1Q	10		MHz
			92																					2Q			
			93																					3Q			
			94																					4Q			
			95																					5Q			
			96																					6Q			
			97																					7Q			
			98																					8Q			

See footnotes at end of device type 05.

MIL-M-38510/37A

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S1 Test no.	1 $\overline{OE}$	2 1D	3 2D	4 3D	5 4D	6 5D	7 6D	8 7D	9 8D	10 GND	11 CLK	12 $9\overline{Q}$	13 $7\overline{Q}$	14 $5\overline{Q}$	15 $5\overline{Q}$	16 $4\overline{Q}$	17 $3\overline{Q}$	18 $2\overline{Q}$	19 $1\overline{Q}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
T_C = 25°C	t_{PLH2}	3003 Fig. 5	99	GND	1H								GND	1H									5.0 V	CLK to 1$\overline{Q}$	4	12	ns
			100	"		1H							"	"							OUT	OUT	"	CLK to 2$\overline{Q}$	"	"	"
			101	"			1H						"	"						OUT			"	CLK to 3$\overline{Q}$	"	"	"
			102	"				1H					"	"					OUT				"	CLK to 4$\overline{Q}$	"	"	"
			103	"					1H				"	"				OUT					"	CLK to 5$\overline{Q}$	"	"	"
			104	"						1H			"	"			OUT						"	CLK to 6$\overline{Q}$	"	"	"
			105	"							1H		"	"		OUT							"	CLK to 7$\overline{Q}$	"	"	"
			106	"								1H	"	"	OUT								"	CLK to 8$\overline{Q}$	"	"	"
	t_{PHL2}	"	107	"	1H								"	"									"	CLK to 1$\overline{Q}$	"	"	"
		"	108	"		1H							"	"							OUT	OUT	"	CLK to 2$\overline{Q}$	"	"	"
		"	109	"			1H						"	"						OUT			"	CLK to 3$\overline{Q}$	"	"	"
		"	110	"				1H					"	"					OUT				"	CLK to 4$\overline{Q}$	"	"	"
		"	111	"					1H				"	"				OUT					"	CLK to 5$\overline{Q}$	"	"	"
		"	112	"						1H			"	"			OUT						"	CLK to 6$\overline{Q}$	"	"	"
		"	113	"							1H		"	"		OUT							"	CLK to 7$\overline{Q}$	"	"	"
		"	114	"								1H	"	"	OUT								"	CLK to 8$\overline{Q}$	"	"	"
	t_{PZL}	"	115	IN	5.0 V								"	1/									"	$\overline{OE}$ to 1$\overline{Q}$	"	18	"
		"	116	"		5.0 V							"	"							OUT		"	$\overline{OE}$ to 2$\overline{Q}$	"	"	"
		"	117	"			5.0 V						"	"						OUT			"	$\overline{OE}$ to 3$\overline{Q}$	"	"	"
		"	118	"				5.0 V					"	"					OUT				"	$\overline{OE}$ to 4$\overline{Q}$	"	"	"
		"	119	"					5.0 V				"	"				OUT					"	$\overline{OE}$ to 5$\overline{Q}$	"	"	"
		"	120	"						5.0 V			"	"			OUT						"	$\overline{OE}$ to 6$\overline{Q}$	"	"	"
		"	121	"							5.0 V		"	"		OUT							"	$\overline{OE}$ to 7$\overline{Q}$	"	"	"
		"	122	"								5.0 V	"	"	OUT								"	$\overline{OE}$ to 8$\overline{Q}$	"	"	"
	t_{PZH}	"	123	"	GND								"	"									"	$\overline{OE}$ to 1$\overline{Q}$	"	"	"
		"	124	"		GND							"	"							OUT	OUT	"	$\overline{OE}$ to 2$\overline{Q}$	"	"	"
		"	125	"			GND						"	"						OUT			"	$\overline{OE}$ to 3$\overline{Q}$	"	"	"
		"	126	"				GND					"	"					OUT				"	$\overline{OE}$ to 4$\overline{Q}$	"	"	"
		"	127	"					GND				"	"				OUT					"	$\overline{OE}$ to 5$\overline{Q}$	"	"	"
		"	128	"						GND			"	"			OUT						"	$\overline{OE}$ to 6$\overline{Q}$	"	"	"
		"	129	"							GND		"	"		OUT							"	$\overline{OE}$ to 7$\overline{Q}$	"	"	"
		"	130	"								GND	"	"	OUT								"	$\overline{OE}$ to 8$\overline{Q}$	"	"	"
	t_{PLZ}	"	131	"	5.0 V								"	"									"	$\overline{OE}$ to 1$\overline{Q}$	3	8	"
		"	132	"		5.0 V							"	"							OUT	OUT	"	$\overline{OE}$ to 2$\overline{Q}$	"	"	"
		"	133	"			5.0 V						"	"						OUT			"	$\overline{OE}$ to 3$\overline{Q}$	"	"	"
		"	134	"				5.0 V					"	"					OUT				"	$\overline{OE}$ to 4$\overline{Q}$	"	"	"
		"	135	"					5.0 V				"	"				OUT					"	$\overline{OE}$ to 5$\overline{Q}$	"	"	"
		"	136	"						5.0 V			"	"			OUT						"	$\overline{OE}$ to 6$\overline{Q}$	"	"	"
		"	137	"							5.0 V		"	"		OUT							"	$\overline{OE}$ to 7$\overline{Q}$	"	"	"
		"	138	"								5.0 V	"	"	OUT								"	$\overline{OE}$ to 8$\overline{Q}$	"	"	"
	t_{PHZ}	"	139	"	GND								"	"									"	$\overline{OE}$ to 1$\overline{Q}$	2	13	"
		"	140	"		GND							"	"							OUT	OUT	"	$\overline{OE}$ to 2$\overline{Q}$	"	"	"
		"	141	"			GND						"	"						OUT			"	$\overline{OE}$ to 3$\overline{Q}$	"	"	"
		"	142	"				GND					"	"					OUT				"	$\overline{OE}$ to 4$\overline{Q}$	"	"	"
		"	143	"					GND				"	"				OUT					"	$\overline{OE}$ to 5$\overline{Q}$	"	"	"
		"	144	"						GND			"	"			OUT						"	$\overline{OE}$ to 6$\overline{Q}$	"	"	"
		"	145	"							GND		"	"		OUT							"	$\overline{OE}$ to 7$\overline{Q}$	"	"	"
		"	146	"								GND	"	"	OUT								"	$\overline{OE}$ to 8$\overline{Q}$	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, R, S test no.	1 $\overline{SC}$	2 1D	3 2D	4 3D	5 4D	6 SD	7 6D	8 7D	9 $\overline{RD}$	10 GND	11 CLK	12 $\overline{8Q}$	13 $7\overline{Q}$	14 $6\overline{Q}$	15 $5\overline{Q}$	16 $4\overline{Q}$	17 $3\overline{Q}$	18 $2\overline{Q}$	19 $1\overline{Q}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Units
10 T_C = 25°C	f_{MAX}																								30		MHz
	t_{PHL2}																								4	15	ns
	t_{PLH2}																								-	-	-
	t_{PZL}	Same tests and terminal conditions as for subgroup 7, except T_C = +125°C.																							-	21	-
	t_{PZH}																								-	-	-
	t_{PLZ}																								7	10	-
	t_{PHZ}																								3	15	-
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ Apply ___ 3.0 V-5.5 V / ___ 0.0 V pulse prior to test.

2/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

3/ Tests shall be performed in sequence, attributes data only.

4/ Output voltages shall be either: (1) H ≥2.4 V minimum and L ≤0.4 V maximum when using a high-speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high-speed checker single comparator.

5/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

6/ I_{IL} limits shall be as follows:

Test	Min/Max limits in (μA) for circuit A	B	C
I_{IL3}	0/-200	0/-200	0/-200

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.8 V, or open).

Subgroups	Symbol	MIL-STD-883 method	Cases K, L, X	Measured terminal	Min	Max	Limits	Unit
1, 2, 3	I_{CC}	3005	1, 2, 3, 4, 5, 6, 7, 8	20		90		mA
			9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	20		90		
			25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38	20		90		
			39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52	20		90		
1, 2, 3	I_{IH1}	3006	1, 2, 3, 4, 5, 6, 7, 8	101		1.0		mA
				102		1.0		
				103		1.0		
				104		1.0		
				201		1.0		
				202		1.0		
				203		1.0		
				204		1.0		
			9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	101		12		
				102		12		
				103		12		
				104		12		
				201		12		
				202		12		
				203		12		
				204		12		
				RCLK		12		
				RCLK		12		
			25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38	RCLK		18		
				RCLK		18		
			39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52	RCLK		18		
				RCLK		18		
1, 2, 3	I_{IL1}	3009	1, 2, 3, 4, 5, 6, 7, 8	RCLK		-18		mA
				101		-18		
				102		-18		
				103		-18		
				104		-18		
				201		-18		
				202		-18		
				203		-18		
				204		-18		
				RCLK		-18		
				RCLK		-18		
			9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	RCLK		-18		
				RCLK		-18		
			25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38	RCLK		-18		
				RCLK		-18		
			39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52	RCLK		-18		
				RCLK		-18		
1, 2, 3	I_{IL1}	3009	1, 2, 3, 4, 5, 6, 7, 8	RCLK		0.4	4.5 V	mA
				101		0.4		
				102		0.4		
				103		0.4		
				104		0.4		
				201		0.4		
				202		0.4		
				203		0.4		
				204		0.4		
				RCLK		0.4		
				RCLK		0.4		
			9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	RCLK		0.4		
				RCLK		0.4		
			25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38	RCLK		0.4		
				RCLK		0.4		
			39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52	RCLK		0.4		
				RCLK		0.4		

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.7 V, or low ≤ 0.8 V, or open).

Test	Symbol	MIL-STD-883 method	Cases	Limits min	Limits max	Unit
Output high voltage	V_{OH}	3006	1, 2, 3	2.4		V
			4	2.7		
			5	2.4		
			6	2.7		
Output low voltage	V_{OL}	3007	1, 2, 3		0.4	V
			4		0.4	
			5		0.4	
			6		0.4	
Input high current	I_{IH1}	3010	1, 2, 3		1.0	μA
			4		1.0	
			5		1.0	
			6		1.0	
Input high current	I_{IH2}	3010	1, 2, 3		40	μA
			4		40	
			5		40	
			6		40	
Input low current	I_{IL1}	3009	1, 2, 3	-1.0		μA
			4	-1.0		
			5	-1.0		
			6	-1.0		
Input low current	I_{IL2}	3009	1, 2, 3	-40		μA
			4	-40		
			5	-40		
			6	-40		
Output short circuit current	I_{OS}	3011	1, 2, 3	-30	-130	mA
			4	-30	-130	
			5	-30	-130	
			6	-30	-130	
Power supply current	I_{CC}	3005	1, 2, 3		180	mA
			4		180	
			5		180	
			6		180	

1/ Measure one output at a time.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 3 / CERDIP L,K / Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits Min	Limits Max	Units
1 T_C = 25°C	I_{CCH}	3005	105	5.0 V	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	GND	GND	5.0 V	2/									2/	5.5 V	V_{CC}		71	mA
	I_{CCL}	"	106	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	-	GND	-								-	-	V_{CC}		79	"
	I_{CCZ}	"	107	GND	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	-	-	-								-	-	V_{CC}		71	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																														
7 4/ T_C = 25°C	Truth table tests	5014	108	A	B	B	B	B	B	B	B	B	B	B	B	B	GND	A		A	K	K	K	Y	X	X	X	X	X	B	5.0 V	All outputs	5/	5/	
			109	-	-	A	A	A	A	A	A	A	A	A	A	A	-	-		A	L	L	L	L	L	L	L	L	L	A	-	-	-	
			110	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		B	-	-	-	-	-	-	-	-	-	B	-	-	-	
			111	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	H	H	H	H	H	H	H	H	H	-	-	-	-	
			112	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	
			113	-	-	B	B	B	B	B	B	B	B	B	B	B	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	
			114	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	
			115	B	-	-	-	-	-	-	-	-	-	-	-	-	B	-		-	-	-	-	L	-	-	-	-	-	-	-	-	-	
			116	-	-	A	A	A	A	A	A	A	A	A	A	A	-	A		-	-	-	-	-	-	-	-	-	-	-	-	-	-	
			117	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		A	-	-	-	-	-	-	-	-	-	-	-	-	-	
			118	A	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-	H	-	-	-	-	-	-	-	-	-	
			119	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	
			120	-	-	B	B	B	B	B	B	B	B	B	B	B	-	-		B	-	-	-	-	-	-	-	-	-	B	-	-	-	
			121	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	L	L	L	L	H	L	L	H	L	-	-	-	-	
			122	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	H	L	L	L	L	H	L	L	H	-	-	-	-	
			123	B	-	-	-	-	-	-	-	-	-	-	-	-	B	-		-	L	H	L	H	L	L	H	L	L	A	-	-	-	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																														
9 T_C = 25°C	t_{PHL1}	3003 Fig. 5	124	3.5 V	GND	IN	IN	IN	IN	IN	IN	IN	IN	IN	GND	1.5 V	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	5.0 V	101	39		MHz
			125	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	102	-		-
			126	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	103	-		-
			127	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	104	-		-
			128	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	201	-		-
			129	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	202	-		-
			130	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	203	-		-
			131	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	204	-		-
	t_{PHL2}	3003 Fig. 5	132	IN	GND	3.5 V	3.5 V	3.5 V	3.5 V	3.5 V	3.5 V	3.5 V	3.5 V	3.5 V	GND	IN	IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	-	1CLK to 101	5	19	ns
			133	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1CLK to 102	-		-
			134	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1CLK to 103	-		-
			135	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1CLK to 104	-		-
			136	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 201	-		-
			137	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 202	-		-
			138	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 203	-		-
			139	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 204	-		-
	t_{PLH2}		140	3.5 V	GND	IN	IN	IN	IN	IN	IN	IN	IN	IN	GND	3.5 V	2/	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	2/	-	1CLK to 101	6	17	ns
			141	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1CLK to 102	-		-
			142	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1CLK to 103	-		-
			143	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1CLK to 104	-		-
			144	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 201	-		-
			145	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 202	-		-
			146	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 203	-		-
			147	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2CLK to 204	-		-

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06. Continued.
Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.8 V, or open).

Group A subgroups	Symbol	MIL-STD-883 method	Case outlines	Limits		Unit	Terminal conditions																							
				min	max		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
							NC	I_{CLK}	V_{CC}	101	102	103	104	201	202	203	204	I_{CLK}	V_{CC}	Required terminal										
1, 2, 3	V_{OH1}	3006	2	2.4		V																								
			3																											
			4																											
			5																											
	V_{OH2}		2	2.4		V																								
			3																											
			4																											
			5																											
	V_{OL1}		2		0.4	V																								
			3																											
			4																											
			5																											
	V_{OL2}		2		0.4	V																								
			3																											
			4																											
			5																											
1, 2, 3	I_{IH1}	3010	2		1.0	μA																								
			3																											
			4																											
			5																											
	I_{IH2}		2		40	μA																								
			3																											
			4																											
			5																											
	I_{IL1}		2	-1.0		μA																								
			3																											
			4																											
			5																											
	I_{IL2}		2	-1.6		mA																								
			3																											
			4																											
			5																											
1, 2, 3	I_{OS1}	3009	2	-20	-55	mA																								
			3																											
			4																											
			5																											
	I_{OS2}		2	-20	-55	mA																								
			3																											
			4																											
			5																											
1, 2, 3	I_{CCH}	3005	2		100	mA																								
			3																											
			4																											
			5																											
1, 2, 3	I_{CCL}		2		100	mA																								
			3																											
			4																											
			5																											

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case outlines X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits Min	Limits Max	Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24				
			Test no.	$1CC_W$	$1DC$	$1D1$	$1D2$	$1D3$	$1D4$	$2Q1$	$2Q2$	$2Q3$	$2Q4$	$2QC$	GND	2CLR	2CLK	2D4	2D3	2D2	2Q1	1D4	1D3	1D2	1D1	1CLR	V_{CC}				
10	f_{MAX}		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																										70		MHz
	t_{PW1}																												5	22	ns
	t_{PLH2}																												8	15	"
	t_{PW2}																												"	"	"
	t_{PHL}																												"	21	"
	t_{PLH}																												"	"	"
	t_{PLZ}																												3	15	"
	t_{PHZ}																												2	10	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = 55°C.																														

1/ Pins not referenced are N/C.

2/ Apply ___ 3.0 V-5.5 V / ___ 0.0 V pulse prior to test.

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

4/ Tests shall be performed in sequence, attributes data only.

5/ Output voltages shall be either: (1) H ≥2.4 V minimum and L ≤0.4 V maximum when using a high-speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high-speed checker single comparator.

6/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

7/ I_{IL} limits shall be as follows:

Test	MIL/Max limits in (μA) for circuit A	B	C
I_{IL}	D7-2Q1	D7-2Q3	D7-2Q2

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high (≥ 2.0 V), low (≤ 0.8 V), or open).

Subgroups	Symbol	MIL-STD-883 method	Case outline	Terminal number																								Limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Min	Max	
$T_C = 25°C$	I_{IH}	3010		I_{IH}																								2.4		V
				Measured terminal																										
	I_{IL}	3009		I_{IL}																									0.8	V
				Measured terminal																										
	I_{OS}	3011		I_{OS}																								-15	-85	mA
				Measured terminal																										
	I_{CCH}	3005		I_{CCH}																										mA
				Measured terminal																										
	I_{CCL}	3005		I_{CCL}																										mA
				Measured terminal																										
$T_C = 25°C$	I_{IH}	3010		I_{IH}																								2.4		V
				Measured terminal																										
	I_{IL}	3009		I_{IL}																									0.8	V
				Measured terminal																										
	I_{OS}	3011		I_{OS}																								-15	-85	mA
				Measured terminal																										
	I_{CCH}	3005		I_{CCH}																										mA
				Measured terminal																										
	I_{CCL}	3005		I_{CCL}																										mA
				Measured terminal																										

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or low ≤ 0.4 V, or open).

Subgroup	MIL-STD-883 method	Symbol	Limits		Unit	Measured terminal	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
			Min	Max			GND	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	2Y	V_{CC}
1, 2, 3	3011	I_{CCH}		55	mA	101																				5.5 V
						102																				
						103																				
						104																				
						201																				
						202																				
						203																				
						204																				
						FCLK																				2.7 V
						RCLK																				
		I_{CCL}		55	mA	101																				5.5 V
						102																				
						103																				
						104																				
						201																				
						202																				
						203																				
						204																				
						FCLK																				7.0 V
						RCLK																				
1, 2, 3	3011	I_{CCH}		55	mA	101																				5.5 V
						102																				
						103																				
						104																				
						201																				
						202																				
						203																				
						204																				
						FCLK																				7.0 V
						RCLK																				
		I_{CCL}		55	mA	101																				5.5 V
						102																				
						103																				
						104																				
						201																				
						202																				
						203																				
						204																				
						FCLK																				7.0 V
						RCLK																				
1, 2, 3	3011	I_{IH}		1.0	µA	51	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	5.5 V
						52																				
						53																				
						54																				
						55																				
						56																				
						57																				
						58																				
						59																				
						60																				
						61																				
						62																				
						63																				
						64																				
						65																				
						66																				
						FCLK																				7.0 V
						RCLK																				
		I_{IL}		1.0	µA	51	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	5.5 V
						52																				
						53																				
						54																				
						55																				
						56																				
						57																				
						58																				
						59																				
						60																				
						61																				
						62																				
						63																				
						64																				
						65																				
						66																				
						FCLK																				7.0 V
						RCLK																				
1, 2, 3	3011	I_{IH}		1.0	µA	51	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	5.5 V
						52																				
						53																				
						54																				
						55																				
						56																				
						57																				
						58																				
						59																				
						60																				
						61																				
						62																				
						63																				
						64																				
						65																				
						66																				
						FCLK																				7.0 V
						RCLK																				
		I_{IL}		1.0	µA	51	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	5.5 V
						52																				
						53																				
						54																				
						55																				
						56																				
						57																				
						58																				
						59																				
						60																				
						61																				
						62																				
						63																				
						64																				
						65																				
						66																				
						FCLK																				7.0 V
						RCLK																				

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.4 V, or low ≤ 0.4 V, or open).

Subgroups	Symbol	MIL-STD-883 method	Cases D, E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{IH1}	3010	1, 2	5.0 V	GND	GND	GND	GND	GND	GND	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	V_{CC}		25	μA
	I_{IH2}	-	1, 2	GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	V_{CC}		25	μA
	I_{IH3}	-	1, 2	GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	V_{CC}		25	μA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and flip flops are melted.																														
3 Truth table tests T_C = 25°C	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 55°C and flip flops are melted.																														
4 T_C = 25°C	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = -55°C and 55°C.																														
	I_{IH4}		120	5.5 V																											
	I_{IH5}		121		5.5 V																										
	I_{IH6}		122			5.5 V																									
	I_{IH7}		123				5.5 V																								
	I_{IH8}		124					5.5 V																							
	I_{IH9}		125						5.5 V																						
	I_{IH10}		126							5.5 V																					
	I_{IH11}		127								5.5 V																				
	I_{IH12}		128									5.5 V																			
	I_{IH13}		129										5.5 V																		
	I_{IH14}		130											5.5 V																	
	I_{IH15}		131												5.5 V																
	I_{IH16}		132													5.5 V															
	I_{IH17}		133														5.5 V														
	I_{IH18}		134															5.5 V													
	I_{IH19}		135																5.5 V												
	I_{IH20}		136																	5.5 V											
	I_{IH21}		137																		5.5 V										
	I_{IH22}		138																			5.5 V									
	I_{IH23}		139																				5.5 V								
	I_{IH24}		140																					5.5 V							
	I_{IH25}		141																						5.5 V						
	I_{IH26}		142																							5.5 V					
	I_{IH27}		143																								5.5 V				
	I_{IH28}		144																									5.5 V			
	I_{IH29}		145																										5.5 V		
	I_{IH30}		146																											5.5 V	
	I_{IH31}		147																												5.5 V

See footnotes at end of table (sheet 07).

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TABLE III. Group A inspection (for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroups (case)	MIL-STD-883 method	Cases	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits Min	Limits Max	Units
			I_{IH}	I_{IL}	I_{OZ}	101	102	103	104	201	202	203	I_{CC}	301	302	303	304	V_{CLK}	305	306	307	308	V_{CLK}	309	310	V_{CC}				
T_C = 25°C	3003	148	3.5 V		GND	IN												3.5 V					2/	OUT	OUT	5.0 V	ICLK to 100		12	ns
		149																									ICLK to 101			
		150																									ICLK to 102			
		151																									ICLK to 103			
		152																									ICLK to 104			
		153																									ICLK to 201			
		154																									ICLK to 202			
		155																									ICLK to 203			
[Test]		156	3.5 V		IN	5.0 V		5.0 V	5.0 V									3.5 V					2/	OUT	OUT	5.0 V	ICLK to 100		14	ns
		157																									ICLK to 101			
		158																									ICLK to 102			
		159																									ICLK to 103			
		160																									ICLK to 104			
		161																									ICLK to 201			
		162																									ICLK to 202			
		163																									ICLK to 203			
[Test]		164	3.5 V		GND	IN		GND	GND									3.5 V					2/	OUT	OUT	5.0 V	ICLK to 100		16	ns
		165																									ICLK to 101			
		166																									ICLK to 102			
		167																									ICLK to 103			
		168																									ICLK to 104			
		169																									ICLK to 201			
		170																									ICLK to 202			
		171																									ICLK to 203			
[Test]		172	3.5 V		IN	5.0 V		5.0 V	5.0 V									3.5 V					2/	OUT	OUT	5.0 V	ICLK to 100		18	ns
		173																									ICLK to 101			
		174																									ICLK to 102			
		175																									ICLK to 103			
		176																									ICLK to 104			
		177																									ICLK to 201			
		178																									ICLK to 202			
		179																									ICLK to 203			
[Test]		180	3.5 V		IN	GND		GND	GND									3.5 V					2/	OUT	OUT	5.0 V	ICLK to 100		20	ns
		181																									ICLK to 101			
		182																									ICLK to 102			
		183																									ICLK to 103			
		184																									ICLK to 104			
		185																									ICLK to 201			
		186																									ICLK to 202			
		187																									ICLK to 203			

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X	2	3	4	5	6	7	8	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	Measured terminal	Limits		Unit
			Case outline	1	2	3	4	5	6	7	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24					Min	Max	
			Terminal no.	1PR	1DC	101	102	103	104	201	202	203	204	20C	GND	2PR	XCLK	270	207	272	271	274	273	272	271	1CLK	V_{CC}					
10 T_C = 25°C	f_{MAX}																														10		MHz
	t_{PHL1}																														5	27	ns
	t_{PLH2}																														9	35	"
	t_{PHL2}																														-	35	"
	t_{P7L}																														-	21	"
	t_{P24}																														-	21	"
	t_{PL2}																														1	35	"
	t_{P47}																														7	30	"
		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																															
11		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																															

1/ Pins not referenced are N/C.

2/ Apply ___1___ 3.0 V-5.5 V / ___1___ 0.0 V pulse prior to test.

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current limit.

4/ Tests shall be performed in sequence, attributes data only.

5/ Output voltages shall be either: (1) H ≥2.4 V minimum and L ≤0.4 V maximum when using a high-speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

6/ f_{MAX} limit is the frequency of the input pulse. The output frequency shall be one half the input frequency.

7/ I_{IL} limits shall be as follows:

Test	MIL/Max limits in (µA) for circuit		
	A	B	C
I_{IL3}	07 -247	07 -200	07 -200

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to group C inspection for class B devices and shall consist of the tests, conditions, and limits specified in subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent.

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

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g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shalll not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

f_{MAX} - - - - - - - - - - - - Maximum clock frequency

GND- - - - - - - - - - - - - Ground, zero voltage potential

V_{IN}- - - - - - - - - - - - Voltage level at an input terminal

V_{OC}- - - - - - - - - - - - Output clamp voltage

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS74
02	54ALS109
03	54ALS112A
04	54ALS574
05	54ALS576
06	54ALS874
07	54ALS876

6.6 Manufacturers' designations. Manufacturers' circuits included in this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Texas Instruments	Motorola	National Semiconductor Corp.
01	X		
02	X		
03	X		
04	X		
05	X		
06	X		
07	X		

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/37A(USAF)	Microcircuits, Digital, Advanced Low-Power Schottky, TTL, Flip Flops

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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